



# **AISSMS** **INSTITUTE OF INFORMATION TECHNOLOGY** **(IOIT)**



ADDING VALUE TO ENGINEERING  
An Autonomous Institute Affiliated to Savitribai Phule Pune University  
Approved by AICTE, New Delhi and Recognised by Govt. of Maharashtra  
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## **Electronics & Telecommunication Engineering Department**

A

MINI PROJECT on

### **Robo Serve**

**SDG 9 - Industry, Innovation and Infrastructure**

SUBMITTED TO AISSMS'S INSTITUTE OF INFORMATION TECHNOLOGY, PUNE  
IN THE PARTIAL FULFILLMENT OF THIRD YEAR MINI PROJECT

## **BACHELOR OF TECHNOLOGY** **IN** **ELECTRONICS AND TELECOMMUNICATION**

### **UNDER THE GUIDANCE OF**

Mr .D.A. Itole

ACADEMIC YEAR: 2025-2026



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| PRN NO    | STUDENT NAME                  |
|-----------|-------------------------------|
| 23310052  | Jadhav Vedant Dnyandeo        |
| 23310053  | Jagdale Pranay Popat          |
| 23310054  | Jagtap Sanket Kailas          |
| 23310055  | Jagtap Sanket Manoj           |
| 23310056  | Jain Riya Sandip              |
| 23310057  | Kale Shruti Ajit              |
| 24320005  | Kalyani Sakshi Shivanand      |
| 23310058  | Kanade Dakshayani Prashant    |
| 23310059  | Kasale Swaraj Milind          |
| 23310060  | Khedekar Shrushti Vilas       |
| 233100561 | Khobragade Rajeshwari Milind  |
| 23310062  | Khose Pratik Bhausaheb        |
| 23310063  | Kolhe Harshada Kishor         |
| 23310064  | Kulkarni Shrawani Padmakar    |
| 23310065  | Langote Priyadarshani Manohar |
| 23310066  | Lokhande Pushkar Amol         |
| 24320006  | Mulani Mehwish Imtiyaz        |
| 23210038  | Nagre Ruchira Shrikisan       |
| 24320027  | Patil Aditi Tanaji            |
| 24320028  | Pawar Shravani Dhananjay      |
| 24320008  | Pillay Tanvi Sachin           |
| 23310005  | Wattamwar Sanskruti Viraj     |



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## CERTIFICATE

This is to certify that Project Report entitled

### **Robo Serve**

Submitted by

### **T3 Batch of (A) Div**

is the record of bonafied work carried out by them under the supervision of Prof.

**Mr.D.A.Itole** in partial fulfillment of the requirement for the Mini Project (TY Btech Electronics and Telecommunication), as prescribed by AISSMS Institute of Information Technology in the Academic Year 2025-2026. This project report has not been earlier submitted to any other Institute or University for the award of any degree or diploma.

**Mr.D.A.Itole**  
Internal Guide

Department of E&TC Engg.

**Dr. Mousami Vanjale**  
Head of Department

Department of E&TC Engg.

External Examiner

**Dr. P. B. Mane**  
Principal  
AISSMS Institute of Information Technology, Pune.

Date:



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# ABSTRACT

The rapid advancement in robotics and embedded systems has significantly contributed to the development of intelligent service-based automation solutions. This project, titled “Smart Autonomous Serving and Greeting Robot”, focuses on the design and implementation of an interactive, semi-autonomous robotic system capable of performing guest greeting and basic service tasks in a structured indoor environment.

The primary objective of this system is to provide a contactless, efficient, and user-friendly solution for welcoming guests and delivering small items such as water bottles in environments like educational institutions, corporate offices, hospitals, and event spaces. The robot integrates multiple subsystems including navigation, obstacle detection, human interaction, and gesture-based communication to enhance user experience.

The system architecture is based on a dual-controller approach, utilizing Raspberry Pi as the central processing unit for high-level operations and ESP32 for real-time control and sensor interfacing. The navigation mechanism is implemented using an IR sensor-based line-following system, enabling the robot to move along a predefined path with stability and accuracy. For safety and collision avoidance, an ultrasonic sensor (HC-SR04) is employed to detect obstacles within a specific range and halt the robot when necessary.

A key feature of the robot is its human-machine interaction capability, achieved through a combination of visual and audio communication. A TFT display is used to present greeting messages, while a speaker module powered by Google Text-to-Speech (gTTS) generates voice-based greetings, creating an engaging and interactive experience for users. Additionally, the robot is equipped with a servo motor-based robotic arm capable of performing simple gestures enhancing its social acceptability and friendliness.

The system is powered by an 11.1V lithium-polymer battery, with voltage regulation achieved using buck converters to supply appropriate levels to different components such as the Raspberry Pi and servo motors. The mechanical design includes a stable chassis with a storage compartment capable of carrying multiple items, ensuring practical usability.

Experimental results demonstrate that the robot successfully performs all intended operations including line following, obstacle detection, guest greeting, and gesture execution with satisfactory accuracy and reliability. Although minor challenges such as sensor sensitivity, servo calibration, and audio latency were encountered, they were effectively addressed during the development process.

**Keywords** - Smart Robot, Raspberry Pi, ESP32, Line Following, Ultrasonic Sensor, TFT Display, gTTS, Robotic Arm, Human Interaction, Automation.

# CHAPTER 1: INTRODUCTION

## 1.1 Relevance / Significance

**Real-World Applicability** The robot addresses a genuine need in hospitality and service industries where automated guest interaction and beverage delivery can improve efficiency and experience. **Post-Pandemic Relevance** In the post-COVID era, contactless service has become a priority. This robot provides a hygienic, touch-free solution for guest interaction and refreshment delivery. **Affordable Automation** Demonstrates that service automation is achievable at low cost using readily available components, making it accessible for small institutions and startups. **Educational Value** The project integrates multiple engineering domains —robotics, embedded systems, Python programming, and AI (gTTS) — making it highly valuable as an academic learning experience. **Human-Robot Interaction** Contributes to the growing field of HRI by demonstrating how simple gestures and voice greetings can make robots more socially acceptable and user-friendly. **Foundation for Research** Serves as a base prototype for advanced research in autonomous service robots, gesture recognition, and AI-driven hospitality systems.

**Social Impact** Reduces dependency on human staff for repetitive tasks, allowing them to focus on more skilled and meaningful work, thereby improving overall workplace productivity.

## 1.2 Problem Statement

"Design and Development of an Autonomous Beverage-Serving Robot with Robotic Arm Gesture Capability and Guest-Greeting "

In modern hospitality, corporate, and institutional environments, the need for efficient, contactless, and interactive guest management has grown significantly. Traditional methods of welcoming guests and serving refreshments rely entirely on human attendants, which introduces challenges such as inconsistency in service, human resource dependency, and limitations during high-footfall events or health-sensitive situations.

There exists a need for an affordable, compact, and semi-autonomous robotic system that can:

1. **Navigate** toward guests or designated locations autonomously
2. **Transport and deliver** beverages/water bottles without human assistance
3. **Greet guests interactively** through voice or display-based communication
4. **Perform human-like gestures** using an integrated robotic arm, creating a more personable and engaging interaction experience
5. **Libraries** To achieve the above objectives, the system relies

Current commercial solutions are either too expensive, too complex, or not tailored for small-scale institutional use such as college lobbies, small offices, clinics, or event halls.

This project aims to bridge that gap by developing a low-cost, wheeled service robot equipped with a storage compartment for beverages, a greeting interface, and a robotic arm capable of performing welcome gestures making automated hospitality accessible and practical.

## 1.3 Specifications

### 1. Microcontroller / Processing Unit

| Parameter       | Details                                                                          |
|-----------------|----------------------------------------------------------------------------------|
| Main Controller | Raspberry Pi , ESP32                                                             |
| Role            | Central processing — line following logic, motor control, display & audio output |

### 2. Robot Movement System

| Parameter             | Details                      |
|-----------------------|------------------------------|
| Motor Type            | Servo Motors(MG996R)         |
| Navigation Mode       | Line Following               |
| Wheel Configuration   | 2 wheeled differential drive |
| Line Detection Sensor | IR Sensors (array)           |

### 3. Robotic Arm

| Parameter | Details                          |
|-----------|----------------------------------|
| Arm Type  | Servo-based Robotic Arm (MG996R) |
| Actuation | PWM-controlled Servo Motors      |
| Function  | Welcome gesture for guests       |

#### 4 Obstacle Detection

| Parameter   | Details                                              |
|-------------|------------------------------------------------------|
| Sensor Used | Ultrasonic Sensor (HC-SR04)                          |
| Function    | Detects obstacles in path and stops/avoids collision |
| Range       | 2 cm to 400cm                                        |

#### 5. Display & Audio System

| Parameter     | Details                    |
|---------------|----------------------------|
| Display       | TFT Display                |
| Audio         | Speaker for voice greeting |
| Greeting Type | Combined visual message    |

#### 6. Payload / Storage

| Parameter         | Details                           |
|-------------------|-----------------------------------|
| Compartment Type  | Enclosed storage box .            |
| Carrying Capacity | 6 water bottles.                  |
| Material          | Lightweight aluminium sheet 1.2mm |

## 7. Power System

| Parameter          | Details                                                                 |
|--------------------|-------------------------------------------------------------------------|
| Battery            | lithium-polymer 11.1V, 5300mAh.                                         |
| Voltage Regulation | Step-down converter for servo supply and power bank for Raspberry Pi 4B |
| Estimated Runtime  | ~1.5 to 2 hours (depending on load)                                     |

## 8. General Dimensions (Approximate)

| Parameter      | Details                 |
|----------------|-------------------------|
| Height         | ~100cm.                 |
| Base Dimension | ~45x 35 cm              |
| Total Weight   | ~15–20kg (with payload) |

## 1.4 Platform Used

### 1. Operating System

Raspberry Pi OS and Rtos OS running on Raspberry Pi, providing the runtime environment for managing all hardware and software operations of the robot.

### 2. Programming Language

Python 3.x, C++ is the primary language used for writing all control logic including motor movement, sensor reading, servo arm gestures, display output, and audio greeting.

### 3. IDE / Code Editor

Arduino IDE Expressive IDE is used for writing, uploading, and debugging the robot's code, along with serial monitor support for real-time testing.

### 4. Libraries Used

| Library             | Purpose                                 |
|---------------------|-----------------------------------------|
| RPi.GPIO / gpiozero | GPIO & servo motor control              |
| gTTS                | GoogleText-to-Speech for voice greeting |
| RPLCD / smbus       | TFT display message control             |
| time / os           | Timing, delays & audio playback         |

### 5. Hardware Platform

Raspberry Pi acts as the central processing unit, interfacing with all sensors , servo motors (locomotion + robotic arm),TFT display, and speaker via GPIO pins using PWM signals.

Text-to-Speech

gTTS (Google Text-to-Speech) converts pre-written welcome text into spoken audio, played through the onboard speaker module whenever a guest is detected

## 1.5 Advantages

- **Contactless Service** — Delivers beverages and greets guests without human involvement, ideal for hygiene-sensitive environments.
- **Cost Effective** — Built with affordable components like Raspberry Pi and servo motors, making it a budget-friendly solution.
- **Interactive Experience** — TFT display combined with voice greeting via gTTS creates an engaging and personalized welcome for guests.
- **Human-like Gesture** — Servo-based robotic arm performing a wave makes the robot more friendly and approachable.
- **Reliable Navigation** — IR sensor-based line following ensures consistent and predictable movement in structured environments.
- **Obstacle Avoidance** — Ultrasonic sensor detects and avoids unexpected obstacles, ensuring safe operation around people.
- **Good Payload Capacity** — Storage compartment carries 6 water bottles, serving multiple guests in a single trip.
- **Long Battery Life** — 11.1V 5200mAh battery provides up to 2 hours of continuous operation.
- **Scalable** — Easily upgradable with features like face recognition, voice commands, or additional gestures in the future.
- **Reduces Human Workload** — Automates repetitive hospitality tasks, freeing up staff for more important work

## 1.6 Applications

- Corporate Offices — Greets visitors at reception and serves water bottles during meetings and events.
- Hotels & Resorts — Assists in welcoming guests and delivering beverages in lobby or waiting areas.
- Hospitals & Clinics — Provides contactless water delivery to patients and visitors, maintaining hygiene.
- College & Universities — Used in seminar halls or conference rooms to serve guests and dignitaries.
- Events & Exhibitions — Engages visitors with interactive greetings and refreshment delivery at stalls or entrances.
- Airports & Railway Stations — Assists travelers by delivering water and providing a welcoming experience.
- Smart Homes — Can be used to serve guests at home in an automated and interactive manner.

# CHAPTER 2: LITERATURE REVIEW

## 2.1 Recent Trends

The field of service robotics has seen rapid advancement in recent years. Some key trends include:

- Autonomous Mobile Robots (AMRs) are increasingly being deployed in hospitals, hotels, and offices for delivery and assistance tasks, replacing manual human effort in repetitive operations.
- Human-Robot Interaction (HRI) has become a major focus, with robots now equipped with voice, gesture, and facial expression capabilities to make interactions more natural and engaging.
- AI & Computer Vision integration using OpenCV and deep learning has enabled robots to recognize faces, detect objects, and navigate dynamically without fixed paths.
- Line Following & Path Planning remain popular in structured indoor environments due to their simplicity, low cost, and high reliability for fixed-route service robots.
- Robotic Arm Gestures are being explored as a means to make service robots more expressive and personable, particularly in hospitality and reception settings.
- Edge Computing on Raspberry Pi has emerged as a cost-effective solution for deploying intelligent robot behavior without relying on cloud infrastructure.
- Robotics-as-a-Service(RaaS)A growing trend is the shift from purchasing robots to subscribing to them. In the RaaS model, companies pay based on usage, reducing high initial costs and making robotics more accessible to small and medium businesses.
- There is increasing research and development in humanoid robots designed to work in human environments such as hospitals and homes. These robots aim to perform complex tasks that require human-like movement and interaction
- Modern robots are integrating advanced AI models (like language and vision systems) to understand environments better and make decisions. This allows robots to perform more adaptive and intelligent tasks in real-world situations

## 2.2 Literature Survey

[1] *Kumar et al. (2021), IEEE* — Proposed an autonomous food delivery robot using Raspberry Pi and IR-based line following. The study demonstrated reliable indoor navigation with obstacle avoidance using ultrasonic sensors, achieving 92% path accuracy in structured environments.

[2] *Sharma & Patel (2022), International Journal of Robotics Research* — Developed a hotel service robot capable of delivering items to rooms using a predefined line path. The robot incorporated a servo-based storage mechanism and an TFTgreeting interface, highlighting the effectiveness of low-cost hardware in hospitality automation.

[3] *Zhang et al. (2021), IEEE Transactions on Human-Machine Systems* — Investigated the role of gesture-based communication in service robots. Results showed that robots with waving or pointing gestures received significantly higher user satisfaction scores compared to non-gestural robots.

[4] *Reddy & Naik (2023), Journal of Embedded Systems* — Implemented a Raspberry Pi-based reception robot with gTTS voice greeting and ultrasonic obstacle detection. The study concluded that voice-enabled greetings improved guest engagement by over 70% compared to display-only systems.

[5] *Liu et al. (2022), IEEE Access* — Explored servo motor-based robotic arms for expressive gestures in social robots. The paper established design guidelines for 2–3 DOF arms suitable for wave and point gestures in human-facing environments.

[6] *Patil & Deshmukh (2023), International Journal of Advanced Engineering* — Presented a beverage-serving robot for college environments using Python on Raspberry Pi. The study emphasized the importance of payload design and battery efficiency in small-scale service robots

## 2.3 Study of System Components

### 1 Raspberry Pi

The Raspberry Pi is a compact, low-cost single-board computer developed by the Raspberry Pi Foundation to promote learning in computing and electronics. Despite its small size, it functions like a full computer, featuring an ARM-based processor, RAM, USB ports, HDMI output, and built-in connectivity such as Wi-Fi and Bluetooth. It typically runs a Linux-based operating system like Raspberry Pi OS and uses a microSD card for storage. One of its key advantages is the presence of GPIO (General Purpose Input/Output) pins, which allow it to interface with sensors, motors, and other electronic components, making it highly suitable for robotics and embedded system applications. Due to its affordability, flexibility, and ease of programming using languages like Python and C, the Raspberry Pi is widely used in projects such as home automation, Internet of Things (IoT) systems, smart devices, and educational



#### 2.3.1 Raspberry pi

### 2 Servo Motors

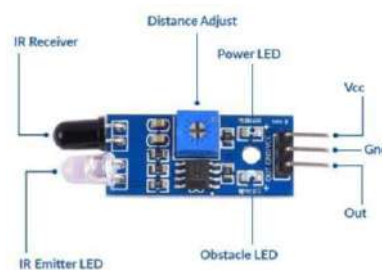
A **Servo Motor** is a type of motor used for precise control of angular or linear position, speed, and movement. It consists of a DC motor, a control circuit, gears, and a feedback system (usually a potentiometer) that continuously monitors the motor's position. Servo motors operate using Pulse Width Modulation (PWM) signals, where the width of the pulse determines the angle of rotation, typically ranging from 0° to 180°. When a signal is sent, the motor rotates to the desired position and holds that position until a new signal is received. These motors are widely used in applications requiring accurate and controlled motion.



### 2.3.2 Servo Motors

## 3 IR Sensors

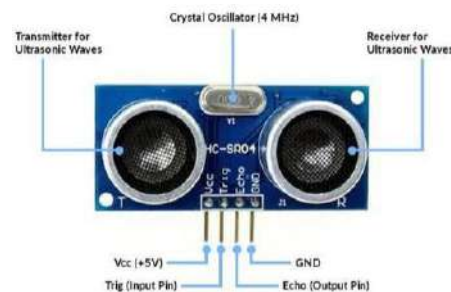
An **Infrared Sensor** is an electronic device used to detect objects, obstacles, or motion by using infrared light. It consists mainly of an IR transmitter (emits infrared radiation) and an IR receiver (detects reflected radiation). When the emitted infrared light hits an object, it reflects back and is captured by the receiver, which then generates a signal indicating the presence of the object. IR sensors work on the principle of reflection and are commonly used for short-range detection. They can operate in both digital (object detected or not) and analog (distance variation) modes. These sensors are widely used in applications such as obstacle detection in robots, line-following robots, automatic doors, and proximity sensing systems. Their advantages include low cost, fast response, simple interfacing, and low power consumption, making them highly suitable for embedded systems and automation projects



### 2.3.3 Ir Sensors

#### 4 Ultrasonic Sensor (HC-SR04)

The HC-SR04 Ultrasonic Sensor is a widely used electronic sensor designed to measure distance using ultrasonic sound waves. It operates by emitting high-frequency sound waves (around 40 kHz) from its transmitter, which travel through the air and reflect back when they hit an object. The sensor then receives the reflected echo through its receiver and calculates the distance based on the time taken for the sound wave to return. This working principle is based on the concept of the Speed of Sound in air. The HC-SR04 module has four pins: VCC (power supply), GND (ground), TRIG (to trigger the ultrasonic pulse), and ECHO (to receive the reflected signal). When a trigger signal is applied, the sensor sends out an ultrasonic burst, and the ECHO pin stays high for a duration proportional to the distance of the object. This time is used in a simple formula to calculate distance.



#### 2.3.4 Ultrasonic Sensor

#### 5 gTTS (Gorq Text-to-Speech)

**gTTS** (often confused as “Gorq Text-to-Speech”) is a Python library used to convert written text into spoken audio using the text-to-speech service of Google. It allows developers to generate natural-sounding speech in multiple languages by simply providing text input in a program. The library works by sending the text to Google’s servers, where it is processed and converted into an audio file (usually in MP3 format), which can then be played on a speaker or stored for later use. It is easy to use and widely applied in projects such as voice assistants, talking robots, accessibility tools, and educational applications. Its key advantages include support for multiple languages, simple implementation in Python, and the ability to produce clear and natural voice output

## 6 TFT Display

A **TFT Display** is a type of liquid crystal display (LCD) that uses thin-film transistor technology to improve image quality, color accuracy, and response time. In a TFT display, each pixel is controlled by its own transistor, which allows for faster switching and sharper images compared to traditional LCDs. It consists of a backlight, liquid crystal layer, and a matrix of transistors that control the intensity of light passing through each pixel to form images. TFT displays are capable of showing high-resolution graphics and a wide range of colors, making them suitable for applications that require clear visual output. They can be interfaced with microcontrollers like Arduino, ESP32, or Raspberry Pi using communication protocols such as SPI or parallel communication. These displays are commonly used in embedded systems, smartphones, medical devices, industrial panels, and robotics projects for displaying data, graphics, and user interfaces. Their main advantages include good brightness, fast response time, compact size, and ease of integration into electronic systems.



### 2.3.6 TFT Display

## 7. 11.1V 5200mAh Battery

An 11.1V 5200mAh Battery is a rechargeable power source commonly used in robotics and electronic projects. The 11.1V rating indicates the battery's output voltage, typically achieved by connecting three lithium-ion cells in series, while the 5200mAh (milliampere-hour) capacity represents how much charge the battery can store, determining how long it can supply power. A higher mAh value means longer backup time. This type of battery provides a stable and efficient power supply for components like motors, microcontrollers, and sensors. It is widely used in robots, drones, and portable electronic systems due to its lightweight design, high energy density, and rechargeability.



### 2.3.7 Battery

## 8. .Motor Driver

A **Motor Driver** is an electronic circuit or module used to control the movement of motors by acting as an interface between a microcontroller (like Arduino, ESP32, or Raspberry Pi) and the motor. Since microcontrollers cannot provide enough current or voltage to drive motors directly, the motor driver takes low-power control signals and converts them into higher-power signals required to run the motor. It can control the direction of rotation (forward/reverse), speed (using PWM signals), and stopping of the motor. Motor drivers are commonly based on H-bridge circuits, which allow bidirectional control of DC motors. Popular motor driver modules include L293D and L298N. These are widely used in robotics, automation systems, and embedded projects where precise motor control is required. Their main advantages include protection of control circuits, efficient power handling, and easy interfacing with different types of motors.



### 2.3.8 Motor Driver

## CHAPTER 3: DESIGN AND DEVELOPMENT



**Robo Serve Images**

## 3.1 Design And Fabrication Of Robot Structure

### 1. Abstract:

This report presents the design and fabrication of the robot structure developed by the Structure & Design Team. The chassis was constructed using a 1.2 mm thick aluminum sheet and reinforced with L-shaped aluminum rods having a cross-section of 2.5 cm × 2.5 cm. The structure was assembled using 3 mm nut-bolts to ensure strong and reliable joints. Additionally, insulated rubber sheets were used within the compartments to provide protection to internal components. The design was developed based on required dimensions and fabricated using standard mechanical processes. The final structure is lightweight, stable, and durable.

### 2. Introduction

The structural design of a robot plays a crucial role in ensuring stability, durability, and proper functioning. A well-designed structure provides mechanical strength and supports all internal components effectively. In this project, the structure was designed using aluminum materials to achieve a balance between strength and weight. Practical considerations such as ease of fabrication, accessibility, and structural rigidity were taken into account during the design process.

### 3. Objectives

- ☐ To design a strong and stable robot structure
- ☐ To achieve lightweight construction
- ☐ To provide proper mechanical support
- ☐ To fabricate the structure using simple tools and methods
- ☐ To maintain dimensional accuracy

### 4. Scope of Work

The work of the Structure & Design Team was limited to the **mechanical design and fabrication** of the robot chassis. This included material selection, structural design, cutting, drilling, and assembly. The interfacing and programming aspects were handled by other teams.

### 5. Materials Used

#### 5.1 Aluminum Sheet

Thickness: 1.2 mm

E & TC, AISSMS IOIT, Pune (centre)

Sheet Size:  $8 \times 4$  ft

Application: Base chassis

Aluminum was selected due to its lightweight nature, corrosion resistance, and ease of fabrication. It provides sufficient strength while maintaining low overall weight.

### 5.2 L-Shaped Aluminum Rods

Cross-section:  $2.5 \text{ cm} \times 2.5 \text{ cm}$

Application: Structural reinforcement

These rods provide rigidity and improve the load-bearing capacity of the structure. Their equal dimensions ensure uniform strength and stability.

### 5.3 Fasteners (Nut and Bolts)

Size: 3 mm

Nut-bolts were used for assembling the structure. They ensure secure joints and allow easy assembly and disassembly when required.

### 5.4 Insulated Rubber Sheet

Insulated rubber sheets were placed inside the compartments of the structure.

Purpose:

- To provide electrical insulation
- To prevent accidental conduction between components
- To avoid damage to internal components
- To reduce vibrations and improve safety

## 7. Structural Design Description

The robot structure is divided into multiple sections to ensure proper support, storage, and stability.

### 7.1 Base Section

The base is a rectangular aluminum platform ( $45 \text{ cm} \times 35 \text{ cm}$ ) that supports the entire structure and provides stability.

### 7.2 Lower Section (Base Compartment)

The lower section has a height of 30 cm and includes an opening at the backside. This opening allows easy access for placing and managing internal components such as battery connections and movement-related hardware.

### 7.3 Middle Section (Storage Compartment)

The middle section, with a height of 40 cm, is a hollow compartment open from the front side. It is designed to store six water bottles.

A specially designed bottle stand is provided to hold the bottles securely and prevent movement during operation.

### 7.4 Neck Section

The section above the middle compartment acts as a connecting structure and is used for storing power banks and necessary cables. This helps in maintaining proper organization inside the structure.

### 7.5 Top Section (Head)

The top section houses the main control setup. It is designed to accommodate essential components such as the Raspberry Pi, display, and cooling fan, while maintaining structural balance and proper ventilation.

### 7.6 Frame Reinforcement

L-shaped aluminum rods are used along edges and joints to provide strength, rigidity, and resistance to deformation.

## 8. Isometric View

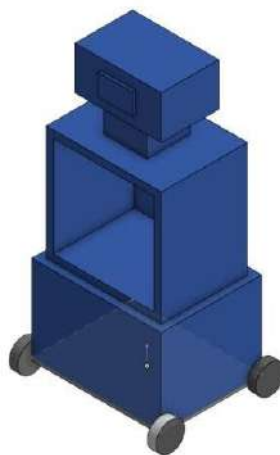


Fig 1 Isometric View

The isometric view provides a three-dimensional representation of the robot structure. It helps in understanding the overall arrangement of different sections, including the base, middle, and top compartments. This view clearly shows how the structure is assembled and how different parts are connected in space.

## 9. Orthographic Views Description

### 9.1 Front View (FV)

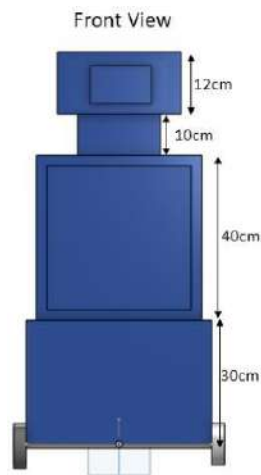


Fig 2 Front View (FV)

The front view shows the vertical structure of the robot, including the height distribution and arrangement of different sections from the front side.

### 9.2 Back View (BV)

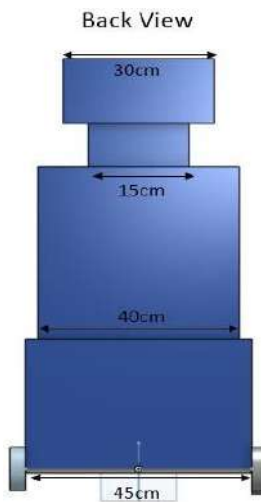


Fig 3 Back View (BV)

The back view represents the rear side of the robot structure and highlights features such as the backside opening and overall symmetry.

### 9.3 Side View (SV)

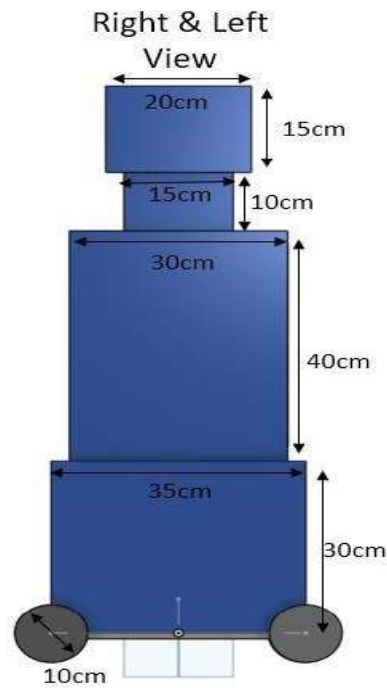


Fig 4 Side View (SV)

The side view illustrates the depth and height of the robot, showing the side profile and vertical alignment of the structure.

### 9.4 Bottom View (BV)

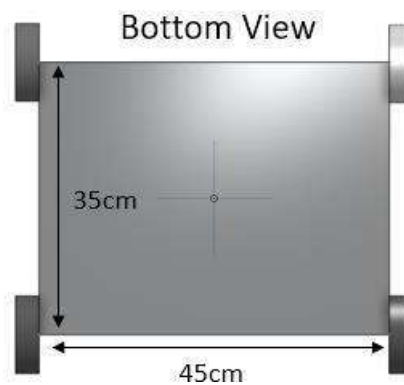


Fig 5 Bottom View (BV)

The bottom view shows the base structure, including the dimensions of the chassis and the arrangement of supporting elements.

## 9.5 Top View (TV)

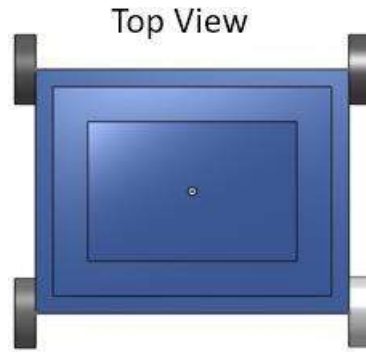


Fig 6 Top View (TV)

The top view represents the layout of the structure from above, showing the width and overall arrangement of the upper sections.

## 10. Design Methodology

The design methodology followed for developing the robot structure was systematic and based on practical requirements.

Initially, the design requirements were carefully analyzed, including size constraints, load-bearing capacity, and overall stability of the robot. Based on these requirements, a preliminary layout of the structure was prepared to visualize the arrangement of different sections.

After finalizing the layout, accurate dimensions were determined for each section of the structure to ensure proper balance and proportional design. Suitable materials such as aluminum sheet and L-shaped rods were selected considering their strength, weight, and ease of fabrication.

Finally, the structure was fabricated and assembled using standard mechanical processes, ensuring that all components were aligned properly and the structure met the required specifications.

## 11. Fabrication Process

The fabrication of the robot structure was carried out using basic mechanical tools and techniques.

First, the required dimensions were carefully measured and marked on the aluminum sheet. The sheet was then cut according to the specified dimensions to form the base and other structural parts.

Holes were drilled at appropriate locations to facilitate assembly and fastening. L-shaped aluminum rods were cut into required lengths to serve as supporting members.

The entire structure was assembled using **3 mm nut-bolts**, ensuring strong and reliable joints. Proper alignment was maintained during assembly to avoid structural imbalance. After assembly, all joints were tightened and the structure was inspected to ensure stability and accuracy.

## 12. Design Considerations

Several important factors were considered during the design of the structure:

- A proper balance between **strength and weight** was maintained to ensure stability without making the robot too heavy.
- The structure was designed to provide sufficient **mechanical strength** to withstand operational loads.
- Ease of fabrication was considered so that the design could be implemented using available tools.
- Accessibility of compartments was ensured by providing front and back openings.
- Insulated rubber sheets were used to provide **electrical insulation and protection** to internal components.
- Adequate spacing and arrangement were maintained for proper organization of internal sections.

## 13. Load and Stress Considerations

The structure was designed considering the expected load and stress conditions during operation.

The aluminum sheet of 1.2 mm thickness was selected to support moderate loads while keeping the structure lightweight. The use of L-shaped rods (2.5 cm × 2.5 cm) significantly improves the **load-bearing capacity** and helps distribute the load evenly across the structure.

The weight of stored items (such as water bottles) and internal components creates vertical loads on the base and middle sections. These loads are transferred through the frame members to the base, ensuring stability.

Stress concentration at joints was minimized by using multiple **3 mm nut-bolts**, which distribute

forces and prevent failure at connection points.

Additionally, the structure was designed to resist:

- **Bending stress** in the aluminum sheet
- **Compressive loads** in vertical members
- **Vibrational effects** during movement

## 14. Challenges Faced

During the fabrication process, several practical challenges were encountered. Maintaining dimensional accuracy was difficult due to manual cutting and marking processes. The thin aluminum sheet tended to bend during handling and fabrication, which required careful attention.

Another challenge was ensuring proper alignment of all structural components during assembly. Even small misalignments could affect the overall stability of the structure.

## 15. Solutions Implemented

To overcome the challenges faced during fabrication, appropriate solutions were implemented.

The structure was reinforced using L-shaped aluminum rods to improve strength and prevent bending. Accurate measurement techniques and careful marking were followed to maintain dimensional precision.

3 mm nut-bolts were used to ensure strong and stable joints. Proper alignment was achieved by assembling the structure step-by-step and checking positioning at each stage.

## 16. Advantages

The developed structure offers the following advantages:

- Lightweight due to the use of aluminum, improving overall mobility
- Sufficient strength to support the required load
- Corrosion-resistant material ensures long-term durability
- Compartment-based design allows organized storage and efficient space utilization
- Insulated rubber sheets enhance safety by preventing electrical conduction and damage
- Easy to assemble and disassemble using nut-bolt connections
- Simple design makes maintenance and modifications convenient
- Provides good structural stability during operation

## **17. Limitations**

Despite its advantages, the structure has certain limitations:

- The 1.2 mm aluminum sheet may bend under excessive load
- Limited load-bearing capacity compared to thicker materials
- Requires careful handling during fabrication and usage
- Manual fabrication may lead to minor dimensional inaccuracies

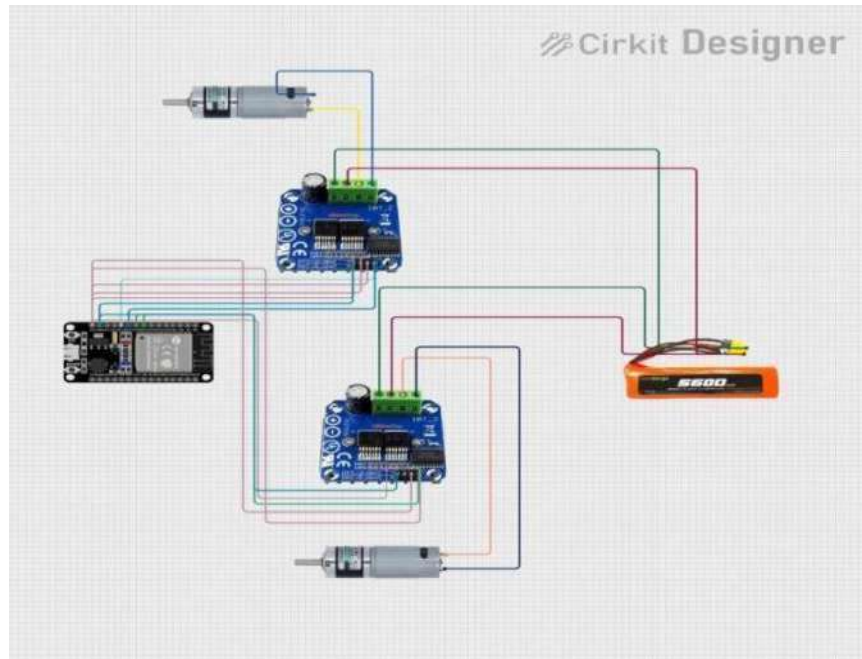
## **18. Future Improvements**

The following improvements can be considered in future:

- Use of thicker aluminum sheet to increase strength and load capacity
- Improvement in finishing for better appearance and durability
- Development of a more compact and optimized design
- Introduction of modular structure for easy upgrades
- Use of advanced fabrication techniques for higher accuracy

## 3.2 Motor Control System

### 1. Circuit Diagram:



### 3.2.1 Motor Control System

## **2. Introduction :**

### Introduction to Motor Control System

The motor control system is a fundamental part of the robotic platform, responsible for converting electrical energy into mechanical motion. In this project, the system consists of an ESP32 microcontroller, BTS7960 motor drivers, planetary DC gear motors, and a LiPo battery.

The ESP32 generates PWM signals to control the speed and direction of the motors through the motor driver. The BTS7960 motor driver acts as an interface between the low-power control signals of the ESP32 and the high-power requirements of the motors. The motors then produce the necessary torque to move the robot, which has a weight of approximately 20 kg.

### Working Principle of Motor Driver System

The ESP32 uses PWM (Pulse Width Modulation) to control motor speed. PWM works by varying the duty cycle of a digital signal, effectively controlling the average voltage supplied to the motor.

Each motor is controlled using two signals:

- RPWM (Right PWM)
- LPWM (Left PWM)

### Direction Control:

- RPWM HIGH, LPWM LOW → Forward
- RPWM LOW, LPWM HIGH → Reverse

- Both LOW → Stop

The BTS7960 motor driver internally uses an H-Bridge configuration, which allows current to flow in both directions through the motor, enabling forward and reverse motion.

### **3. System Specifications**

#### **1. Planetary DC Gear Motor**

The selection of the motor is a critical aspect in the design of the robotic system, as it directly influences the robot's mobility, load-carrying capacity, and stability. In this project, a 40W planetary DC gear motor with 200 RPM speed and 20–25 kg·cm torque has been selected based on the structural and operational requirements.

The robot's weight of 20 kg requires sufficient torque to overcome inertia and rolling resistance. Additionally, due to its height of 100 cm, the center of gravity is relatively high, making smooth acceleration and controlled motion essential to prevent instability.

The planetary gearbox plays a vital role in enhancing motor performance. It distributes load across multiple gears, resulting in:

- Higher torque output
- Improved efficiency
- Compact design
- Increased durability

The selected speed of 200 RPM provides an optimal balance between speed and control. Higher speeds may cause instability, while lower speeds reduce operational efficiency. The 40W power rating ensures



#### **3.2.2 Dc Gear Motor**

## Design Calculations

### 1. Force Acting on the Robot

$$F = m \times g$$

$$F = 20 \times 9.81 = 196.2 \text{ N}$$

Explanation:

This represents the total gravitational force acting on the robot.

### 2 Rolling Resistance Force

$$F_r = \mu \times m \times g$$

$$F_r = 0.02 \times 20 \times 9.81 = 3.924 \text{ N}$$

Explanation:

This is the actual force required to move the robot on a surface, considering rolling resistance.

### 3 Torque Required at Wheel

$$T = F_r \times r$$

$$T = 3.924 \times 0.05 = 0.1962 \text{ Nm}$$

Explanation:

This is the torque required at the wheel to move the robot.

### 4 Torque per Motor

$$T \text{ per motor} = T / 2$$

$$T \text{ per motor} = 0.1962 / 2 = 0.0981 \text{ Nm}$$

Explanation:

Since two motors are used, the total torque is divided equally.

### 5 Motor Torque Conversion

$$1 \text{ kg}\cdot\text{cm} = 0.0981 \text{ Nm}$$

$$\text{Motor torque} = 20 \times 0.0981 = 1.962 \text{ Nm}$$

Explanation:

Motor torque is converted into SI unit (Newton-meter) for comparison.

### 6 Safety Factor

$$\text{Safety Factor} = \text{Motor Torque} / \text{Required Torque}$$

Safety Factor =  $1.962 / 0.0981 \approx 20$

Explanation:

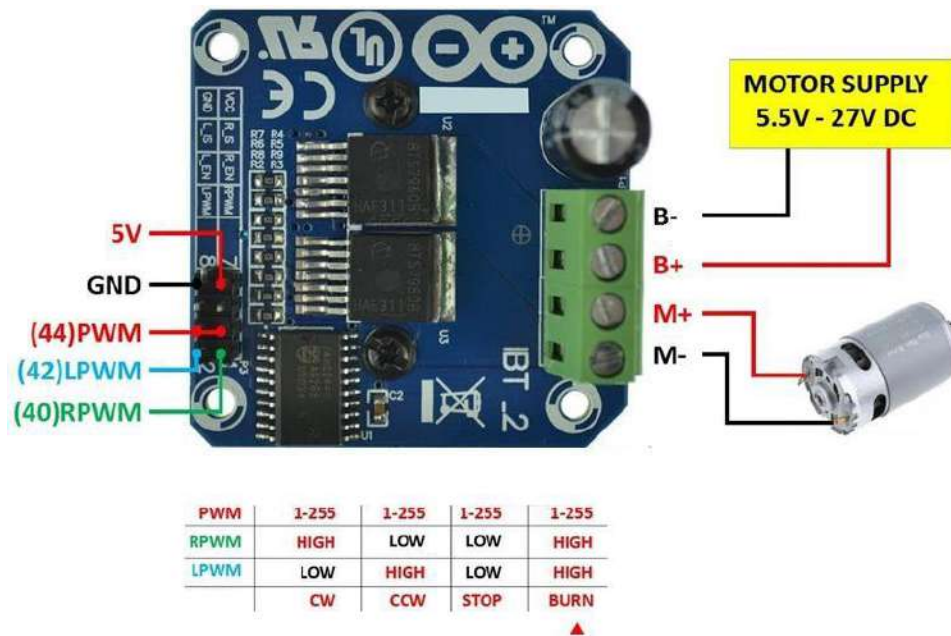
Motor provides approximately 20 times more torque than required

## **2. Motor Driver (BTS7960)**

The selection of the motor driver is a critical aspect in the design of the robotic system, as it directly affects the control, efficiency, and safe operation of the motors. In this project, the BTS7960 motor driver has been selected due to its high current handling capability and compatibility with high-power DC gear motors used in the system.

The robot utilizes 40W DC gear motors that require significant current, especially during starting and load conditions. Since the ESP32 microcontroller operates at low voltage and cannot supply high current directly, a motor driver is essential to amplify the control signals and drive the motors effectively. The BTS7960 driver is capable of handling currents up to 43A, making it suitable for the given application.

The motor driver operates based on the H-Bridge principle, which allows bidirectional control of the motor by controlling the direction of current flow. This enables forward and reverse motion of the robot. The ESP32 provides PWM signals to the motor driver, allowing precise control of motor speed and direction.



### 3.2.2 Motor Driver

The BTS7960 motor driver provides several advantages that make it suitable for this robotic system:

- High current handling capability
- Efficient PWM-based speed control
- Bidirectional motor control using H-Bridge
- Built-in protection against overcurrent and overheating

Due to these features, the BTS7960 ensures reliable motor operation, smooth speed control, and protection of the control circuitry. Hence, it is an appropriate choice for driving high torque motors in the proposed robotic system.

### 3. Power Supply System (Battery)

The selection of the battery is a critical aspect in the design of the robotic system, as it directly affects the robot's runtime, performance, and reliability. In this project, a 11.1V, 5200mAh, 40C Lithium Polymer (LiPo) battery has been selected based on the power requirements of the motors, motor driver, and control system.

The robot uses high torque DC gear motors which require significant current, especially during startup and load conditions. Therefore, the battery must be capable of supplying both stable voltage and high current without voltage drop. The selected battery provides a nominal voltage of 11.1V, which is suitable for efficient motor operation.

The capacity of 5200mAh ensures sufficient runtime for continuous operation. Additionally, the 40C discharge rating allows the battery to deliver high current when required. The maximum current that can be supplied is calculated as:

$$I_{\text{max}} = \text{Capacity} \times \text{C-rating}$$

$$I_{\text{max}} = 5.2 \times 40 = 208 \text{ A}$$

This ensures that the system can handle peak current demands without performance degradation.



### 3.2.3 Battery

The Lithium Polymer battery offers several advantages that make it suitable for this robotic system:

- High energy density
- Lightweight and compact design
- High current discharge capability
- Stable and rechargeable power source

Due to these features, the selected battery ensures reliable and efficient power delivery to the robotic system.

## 4. Controller and Communication System (ESP32 with RTOS and UART)

The controller plays a vital role in the operation of the robotic system, as it is responsible for processing sensor data and controlling motor movement. In this project, the ESP32 microcontroller is used as the main controller due to its high processing capability, built-in peripherals, and support for real-time applications.

The ESP32 operates using a Real-Time Operating System (RTOS), which allows multitasking and efficient management of different operations such as sensor reading, motor control, and communication. RTOS ensures that time-critical tasks are executed without delay, improving the overall performance and responsiveness of the robot.

The system is divided into multiple tasks, such as:

- Motor control task
- Sensor processing task
- Communication task

This task-based execution allows parallel processing and ensures smooth operation of the robot.

For communication, UART (Universal Asynchronous Receiver Transmitter) is used to establish a connection between the ESP32 and the Raspberry Pi. The Raspberry Pi performs high-level tasks such as face recognition and AI-based interaction, while the ESP32 handles real-time motor control.



#### 3.2.4 ESP 32

The communication works as follows:

- Raspberry Pi sends command signals (e.g., start, stop, direction)
- ESP32 receives the data via UART
- Based on the received command, ESP32 controls the motors

This separation of tasks improves system efficiency by dividing computational load between the two devices.

The ESP32 provides several advantages for this robotic system:

- Dual-core processor for parallel execution
- RTOS support for real-time task management
- Built-in UART communication
- Low power consumption
- High processing speed

Due to these features, the ESP32 ensures efficient control, fast response, and reliable communication within the robotic system.

## 3.3 Line Follower

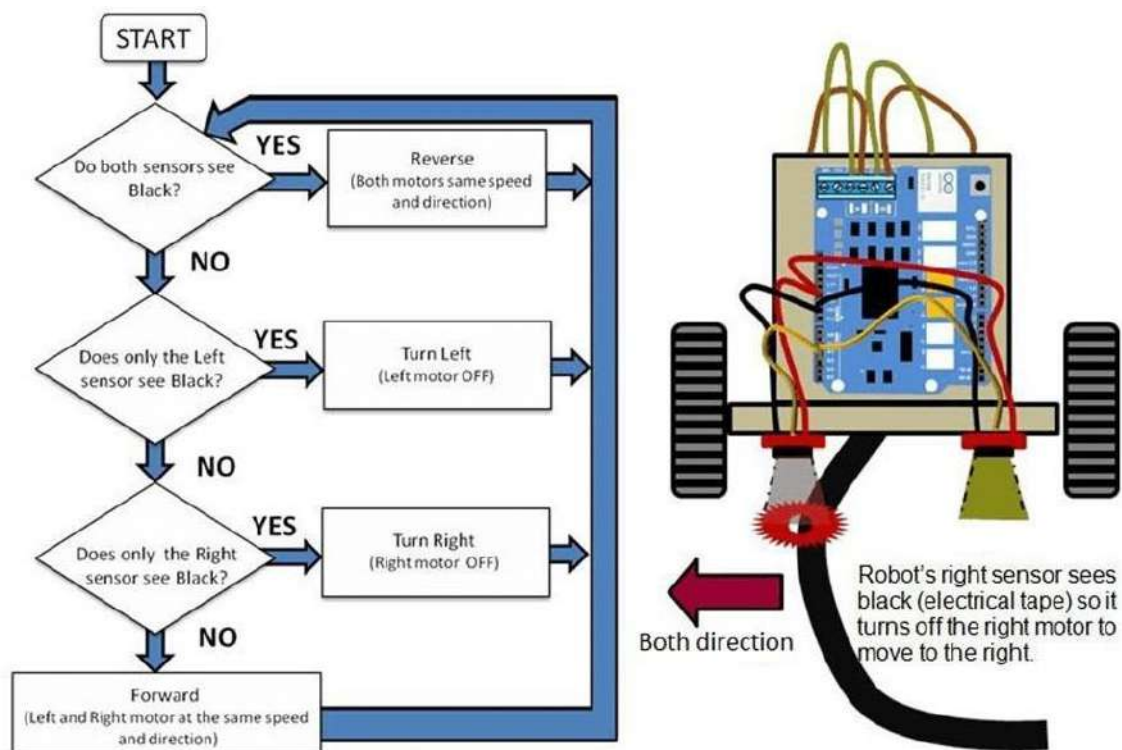
### 1. Introduction

A **line follower robot** is an autonomous robotic system designed to follow a predefined path, typically a black line on a white surface or vice versa. It operates using sensors to detect the path and a control system to guide its movement accordingly.

In this project, an **8-channel infrared (IR) sensor array** is used to achieve high accuracy in line detection. Unlike basic line followers that use only 2 or 3 sensors, the use of multiple sensors allows the robot to precisely determine the position of the line relative to its body. This results in smoother motion, better handling of curves, and improved performance at intersections.

In this project, an **8-channel IR sensor array** is interfaced with the ESP32 microcontroller to detect the position of a line. Each sensor provides a digital output indicating whether it is over a black line or a white surface. These outputs are read by the ESP32 and used to control the movement of the robot.

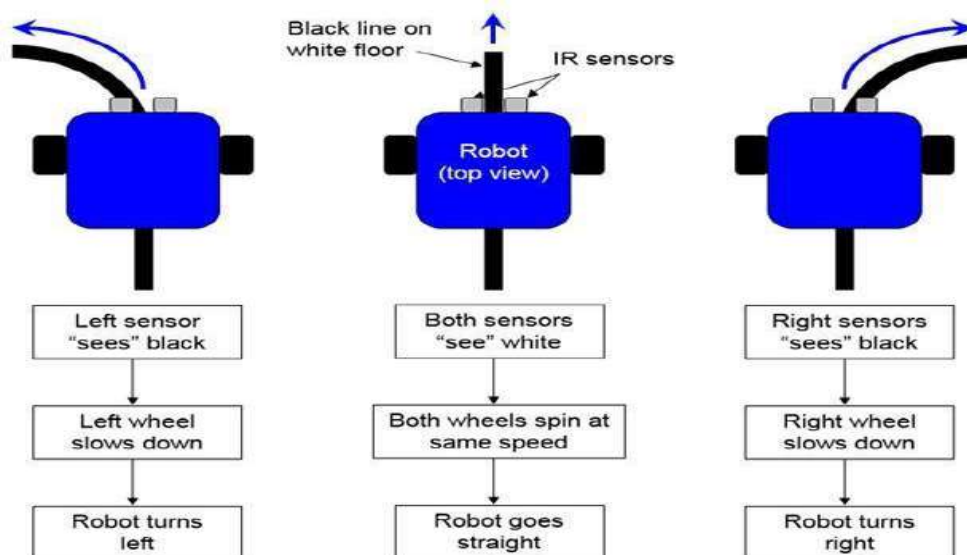
### 2. Flow Diagram



#### 3.3.1 line follower Flow Diagram

### 3. Working of line follower

The line follower robot operates using an 8-channel IR sensor array connected to an ESP32 microcontroller, which continuously detects the position of a line based on infrared light reflection. When the sensors pass over a black line, they output a LOW signal, and when over a white surface, they output HIGH. The ESP32 reads these signals in real time and determines the position of the line relative to the robot. If the central sensors detect the line, the robot moves forward with both motors running at equal speed. If the line shifts slightly to the left or right, the ESP32 adjusts the motor speeds to turn the robot smoothly in the required direction. For sharp deviations, the robot performs pivot turns by rotating the motors in opposite directions. In cases where the line is not detected by any sensor, the robot uses a memory-based approach by referring to its last turning direction to search for the line. This continuous process of sensing, decision-making, and motor control enables the robot to accurately follow the path in real time.



The 8-channel IR sensor array works on the principle of **infrared light reflection**. Each sensor consists of two main components:

- **IR LED (Emitter)** → emits infrared light
- **Photodiode/Phototransistor (Receiver)** → detects reflected light

#### ◆ Working Mechanism

When the sensor is placed near a surface:

- On a **white surface**:
  - More IR light is reflected
  - Receiver detects strong signal
  - Output = **HIGH (1)**
- On a **black surface (line)**:

- IR light is absorbed

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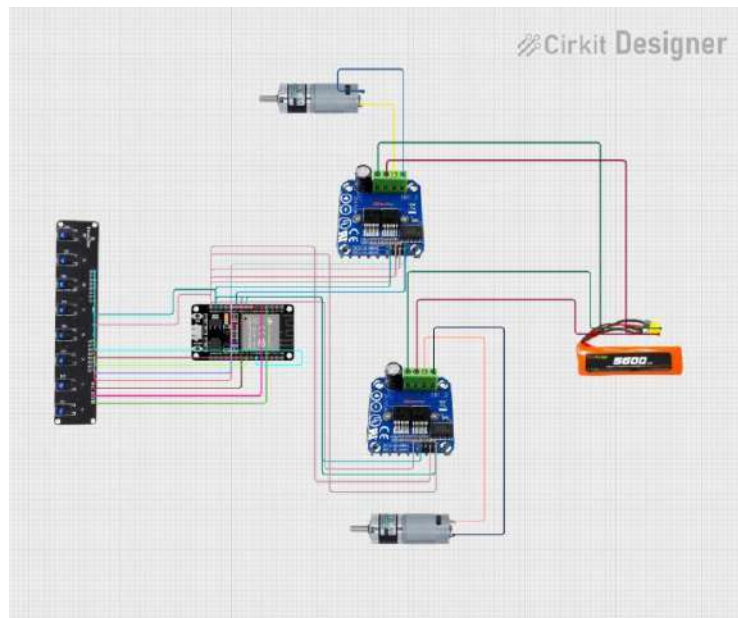
- Very little reflection
- Output = **LOW (0)**

#### ◆ Role of Comparator (LM393)

Each sensor module typically uses an **LM393 comparator**:

- Converts analog signal → digital output
- Compares reflected light with a threshold
- Outputs clear HIGH or LOW signal

#### 4. Connection Diagram



#### 3.3.2 Line Follower Circuit Diagram

#### 5. Component Specifications

- **ESP32:**  
Operates at 3.3V, supports multiple GPIO pins and PWM channels, used for processing sensor data and controlling motors.
- **BTS7960 Motor Driver:**  
Works at 5V–27V, supports high current up to 43A, used to control speed and direction of DC motors using PWM signals.
- **8-Channel IR Sensor Array:**  
Operates at 3.3V–5V, gives digital output (0/1) based on black/white detection using infrared reflection.
- **DC Motors:**  
12V geared motors (~200 RPM), provide movement to the robot.

- Battery:  
12V rechargeable battery, supplies power to motors and driver.
- Chassis:  
Mechanical base to hold all components.

This circuit represents a **line-following robot system** built using an ESP32 microcontroller, an 8-channel IR sensor array, two IBT-2 motor drivers, and two DC geared motors powered by a battery. The battery supplies power mainly to the motor drivers, which require higher current to drive the motors, while the ESP32 operates at a lower regulated voltage. A common ground is maintained across all components to ensure proper signal reference and stable operation. The IR sensor array continuously detects the surface beneath it by emitting infrared light and sensing the reflection; white surfaces reflect more light (logic HIGH), while black lines absorb light (logic LOW). These digital signals are sent to the ESP32, which acts as the brain of the system.

The ESP32 reads the sensor values in real time and determines the position of the line relative to the robot. Based on this input, it decides how the motors should move. For example, if the center sensors detect the line, the robot moves straight; if the line shifts toward the left sensors, the robot turns left; and if it shifts toward the right sensors, it turns right. These decisions are translated into PWM (Pulse Width Modulation) signals sent to the IBT-2 motor drivers. Each motor driver controls one DC motor and uses PWM inputs to regulate both the speed and direction of rotation. By adjusting the speed of the left and right motors independently, the robot can smoothly follow the path. Overall, the system works in a continuous loop of sensing, processing, and actuation, enabling the robot to autonomously follow a line path.

## 5.MATHEMATICAL MODELLING & CONTROL LOGIC:

Let,

$S_L$  = Left Sensor Output

$S_R$  = Right Sensor Output

$e$  = error signal

$V_L$  = Left Motor Speed

$V_R$  = Right Motor Speed

$V_{Base}$  = constant base Speed

$V_{high}, V_{low}$  = Predefined Motor Speed

Sensor Logic :

1 for Line detected(black)

0 for Surface Detected (white)

The System Error is defined as,

$$e = S_L - S_R$$

Thus :

$e = 0$  : Robot is Centered on line

$e = +1$  : Left Deviation

$e = -1$  : Right Deviation

### Bang-Bang Control Law

$$(V_L, V_R) = \begin{cases} (V_{base}, V_{base}), & \text{if } e = 0 \\ (V_{low}, V_{high}), & \text{if } e = +1 \\ (V_{high}, V_{low}), & \text{if } e = -1 \end{cases}$$

Control Logic :

In bang-bang control, the system does not apply gradual correction as in PID. Instead it Switches predefined motor speed based on sensor input . When both sensor detect the line, the robot moves straight , when left detect and right does not robot turns right and vice-versa.

System behavior Analysis :

$e = 0$  : Robot is aligned move forward with equal speed

$e = +1$ : Left Sensor is active , Robot steers right

$e = -1$ : Right Sensor is active, Robot steers left

The control decision is governed by following :

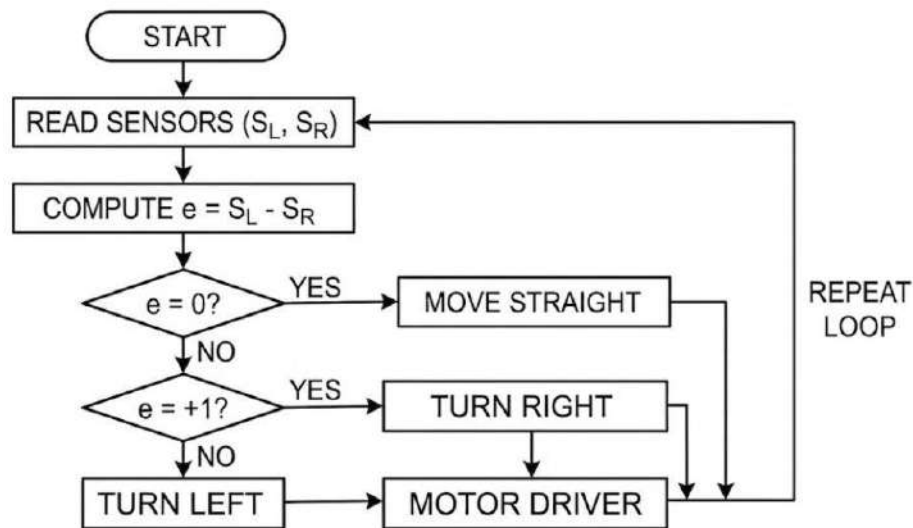
If  $S_L = S_R$  = Move Straight

If  $S_L > S_R$  = Turn Right

If  $S_L = S_R$  = Move Straight

## 6. Algorithm / Control Logic :

1. Initialize the system and set motor speed to base values
2. Continuously read sensor values  $S_L$  and  $S_R$ .
3. Compute error using:  $e = S_L - S_R$ .  
If  $e = 0$ : Both motors run at base speed  $\rightarrow$  Robot moves straight.  
If  $e = +1$ : Left motor speed is reduced and right motor speed is increased  $\rightarrow$  Robot turns right.  
If  $e = -1$ : Left motor speed is increased and right motor speed is reduced  $\rightarrow$  Robot turns left.  
Repeat the above steps continuously for real-time correction.



### 3.3.3 Line Follower Algorithm

## 7. Limitations :

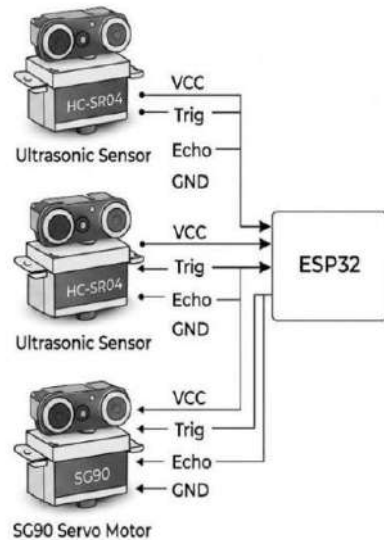
- The system exhibits oscillatory behavior near the line due to abrupt switching of control signals.
- Sensor readings may vary due to ambient light conditions.
- Sharp turns are difficult to handle efficiently using bang-bang control.
- Motor response delay affects accuracy.

## 8. Future Implementations :

- Implementation of PID control for smoother motion.
- Use of multiple sensors for better line detection.
- Filtering techniques to reduce sensor noise.
- PWM-based speed control instead of discrete switching.

### 3.4 Obstacle Detection System:

#### 1. Circuit Diagram:



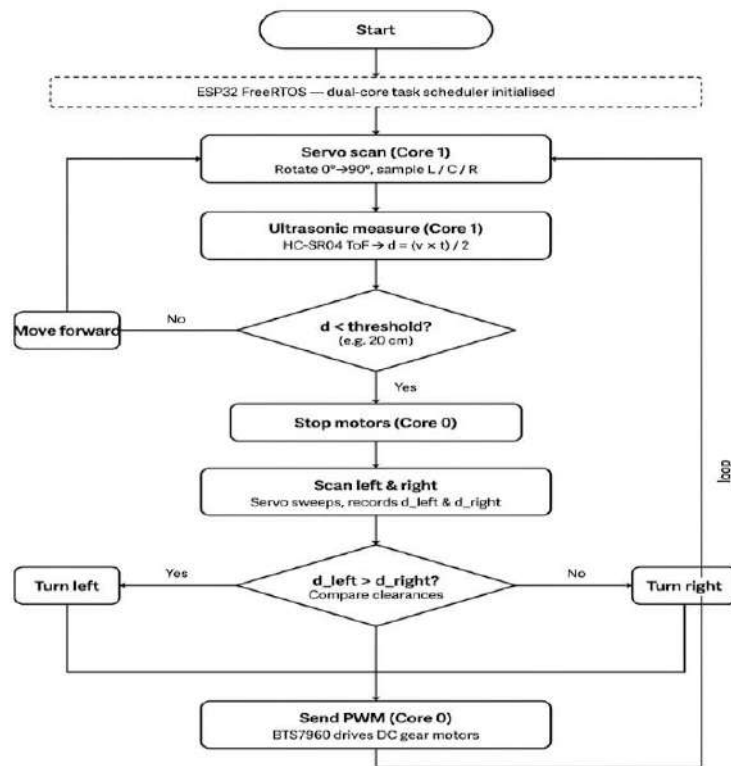
#### 2. Introduction

The ultrasonic-based obstacle avoidance system is a critical subsystem of the robotic platform, responsible for detecting obstacles in the surrounding environment and enabling safe navigation. In this project, the system consists of an ESP32 microcontroller, ultrasonic sensor (HC-SR04) Time Of Flight sensor, Servo motor, BTS7960 motor drivers, planetary DC gear motors, and a LiPo battery.

The ultrasonic sensor detects obstacles by transmitting high-frequency sound waves and measuring the time taken for the echo to return. The ESP32 processes this data and determines the distance of objects in front of the robot. To improve environmental awareness, the ultrasonic sensor is mounted on a servo motor, which rotates across a predefined angular range. This arrangement creates a radar-like scanning system that detects obstacles in multiple directions (left, front, right).

The ESP32 uses its dual-core architecture with FreeRTOS to efficiently manage tasks. Core 1 is dedicated to real-time obstacle detection and servo scanning, while Core 0 handles motor control and communication. Based on sensor data, the ESP32 controls the motors via the BTS7960 motor driver to stop or change direction when an obstacle is detected.

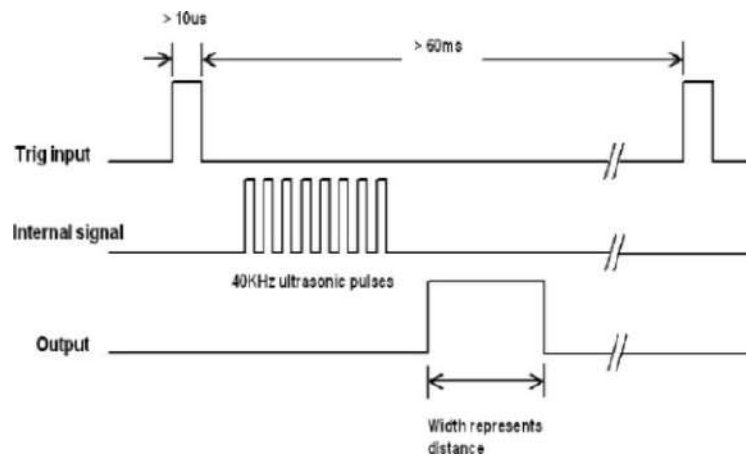
### 3. System Flow Diagram:



#### 3.4.1 Obstacle Detection System

### 4. Working Principle of Ultrasonic Detection System

The ultrasonic sensor operates based on the time-of-flight principle. The sensor transmits ultrasonic waves at a frequency of 40 kHz. When these waves encounter an obstacle, they are reflected back and received by the sensor. The time difference between transmission and reception is used to calculate the distance.



The distance is calculated using:

$$d = \frac{v \times t}{2}$$

Where:

- $d$  = distance to obstacle
- $v$  = speed of sound (343 m/s)
- $t$  = time for echo

If the measured distance is less than a predefined threshold (e.g., 20 cm), the robot stops to prevent collision.

### **Servo-Based Radar Scanning System**

The servo motor is used to rotate the ultrasonic sensor to scan multiple directions. The scanning mechanism works as follows:

- Servo rotates from 0° to 90° to avoid wrong readings
- Distance is measured at different angles
- Data is collected for left, center, and right directions
- A simple environmental map is created

#### **5. Advantages of Servo-Based Scanning**

- Wider field of view
- Improved obstacle detection accuracy
- Direction-based navigation
- Efficient path selection

#### **6. Obstacle Avoidance Algorithm**

The robot follows a decision-based algorithm for navigation:

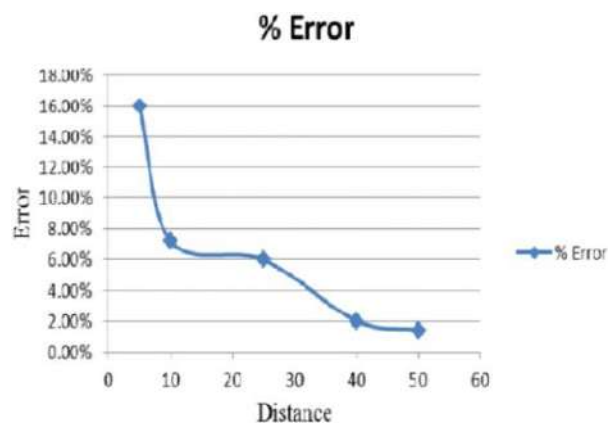
1. Move forward
2. Measure distance using ultrasonic sensor
3. If distance > threshold → continue forward
4. If distance < threshold:

- Stop the robot
- Rotate servo to left → measure distance
- Rotate servo to right → measure distance
- Compare distances
- Move in the direction with maximum clearance

This ensures intelligent and collision-free navigation.

### **Total Area Covered :**

To calculate area covered we need to see the error graph as follows



We can see that as the distance increases the error in reading smoothens out so it is pretty unreliable for proximity sensing but nevertheless can be used.

The total area covered is as follows :

Let,

$A$  = Area covered

$R$  = max sensing range (20 cm )

$\theta$  = angle sweep by servo

so area is  $A = 1/2(R)^2 * \theta$

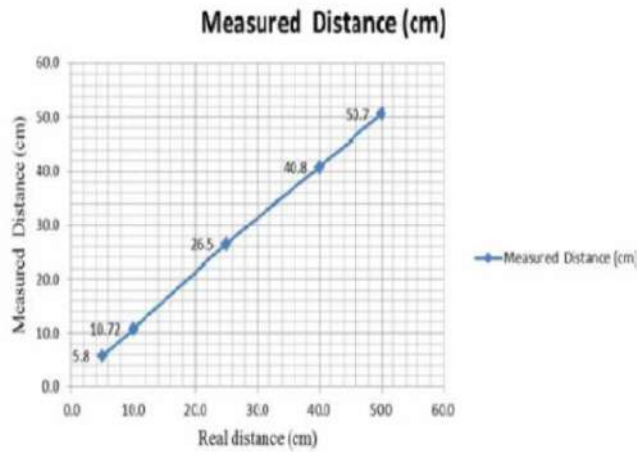
for the front facing servo : the angle swept is 180 Degrees ,

so Area covered is  $A = 3.6 \text{ m}^2\text{-rad}$ .

And for the side servo's which have capability of going to 90 degree

$A = 1.8 \text{ m}^2\text{-rad}$ .

Following is the graph representing the accuracy :



## ESP32 Dual-Core Processing with FreeRTOS

The ESP32 microcontroller uses a dual-core architecture to improve system performance and real-time response.

### Core Allocation

Core 1 (Real-Time Processing):

- Ultrasonic sensor reading
- Obstacle detection
- Servo motor scanning

Core 0:

- Motor control
- UART communication
- Debugging and monitoring

FreeRTOS is used to divide the system into multiple tasks that run simultaneously.

### Task Distribution

- Sensor Task – reads distance data
- Servo Task – performs scanning
- Motor Control Task – controls movement
- Communication Task – handles UART

This parallel execution ensures fast response and efficient operation.

## Design Considerations

The design of the ultrasonic-based obstacle avoidance system requires careful selection of components and system architecture to ensure reliable and efficient operation. We are using 3 Ultrasonic sensor in

a triad fashion ie one up front and two on the two sides which are left and right, initially it was decided to use 6 ultrasonic sensor but since the ultrasonic sensor are unreliable we choose 3 . When the ultrasonic transmit it bombards the space with sound waves and since they are omni-directional the other sensors data gets corrupted to avoid this we are using 3 ultrasonic sensor mounted on servo motor to get maximum area covered.

### **1. Sensor Placement**

The ultrasonic sensor is mounted on a servo motor at an elevated position to avoid ground reflections and ensure accurate readings. The height and angle are selected to maximize detection range and minimize noise.

### **2. Scanning Angle Limitation**

The servo is operated between  $0^{\circ}$  to  $90^{\circ}$  instead of full  $180^{\circ}$  to Reduce scanning time, avoid inaccurate side reflections, Improve response speed.

### **3. Threshold Distance Selection**

A threshold distance of 20 cm is chosen based on: Robot speed, Stopping distance, Sensor accuracy. This ensures timely stopping without false triggering.

### **4. Real-Time Processing**

Core 1 of ESP32 is dedicated to obstacle detection to ensure:

- Faster response
- No delay due to communication tasks
- Improved safety

Free RTOS is used to reduce timing problems and ensure absolute real-time operation.

### **5. Power Management**

Separate power regulation is used for:

- Motors (high current)

Since the motors require peak current of 23 A and burst current 27A , we are using 5200mAh 11.1V Li-Po battery with 40C rating so it can provide Theoretically 208A but considering losses in contact and electrical components we can expect a decline of 20-30A hauling the current output to a max of 178A.

- ESP32 and sensors (low power via buck down)

The ESP32 has a voltage rail of 5V and 3V but the extensa LX6 core uses 3.3V so to ensure max performance were using the 5V rail . The sensors , servo MG96 , IR array and HC-SR04 operate at optimal operating voltage of 5V . So considering the requirement of both the systems we are using a buck down that bucks down the 12V battery to 5V and it is given to 5V rail of ESP32 .This avoids voltage fluctuations and system instability.

## 6. Noise and Interference Handling

To improve accuracy:

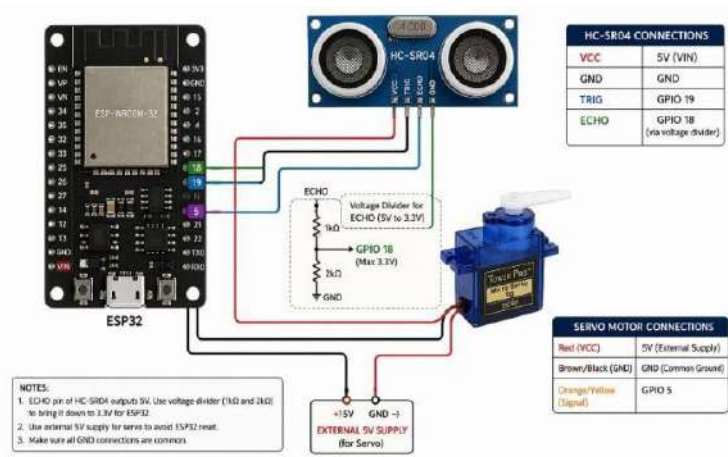
- Multiple readings are averaged
- Delay is added between readings
- Sudden spikes are filtered

## 7. Working Principle of Integrated System

The complete system works as an integration of sensing, processing, and actuation:

1. Ultrasonic sensor measures distance
2. ESP32 processes data (Core 1)
3. If obstacle detected:
  - Motor stops
  - Servo scans environment
4. Best direction is selected
5. Motor driver receives PWM signals
6. Robot moves safely

## 8. Circuit Diagram:



### 3.4.1 Obstacle Detection System Circuit Diagram

## **9. System Specifications:**

The system specifications define the electrical, mechanical, and functional characteristics of the components used in the ultrasonic-based obstacle avoidance robot. Proper selection of these specifications ensures reliable operation, efficient performance, and system stability.

### **1. Ultrasonic Sensor (HC-SR04)**

The HC-SR04 ultrasonic sensor is used for distance measurement based on the acoustic time-of-flight principle. It operates at a frequency of 40 kHz, which is beyond the audible range of humans. The sensor transmits ultrasonic pulses and measures the time taken for the reflected signal to return after hitting an obstacle.

The sensor operates at a voltage of 5V and provides a detection range from 2 cm to 400 cm, making it suitable for short- to medium-range obstacle detection. It offers an accuracy of approximately  $\pm 3$  mm under ideal conditions. However, environmental factors such as temperature, humidity, and surface texture of the obstacle may affect measurement accuracy.

### **2. Servo Motor**

The servo motor is used to provide rotational motion to the ultrasonic sensor for scanning purposes. It operates at 5V and is controlled using Pulse Width Modulation (PWM) signals generated by the ESP32.

The servo motor supports a rotation range of  $0^\circ$  to  $180^\circ$ , but in this system, it is typically limited to  $0^\circ$  to  $90^\circ$  to improve scanning speed and reduce erroneous readings. The servo enables directional sensing by rotating the sensor to different angles, thereby creating a radar-like scanning mechanism.

### **3. ESP32 Microcontroller**

The ESP32 serves as the main controller of the system and is responsible for processing sensor data, executing control algorithms, and managing communication. It features a dual-core processor with a clock speed of up to 240 MHz, enabling high-speed computation and multitasking.

The ESP32 supports a Real-Time Operating System (FreeRTOS), which allows the system to be divided into multiple concurrent tasks such as sensor processing, servo control, motor control, and communication. It also includes built-in WiFi and Bluetooth capabilities, which can be used for wireless monitoring and control in advanced applications.

#### **4. Motor Driver (BTS7960)**

The BTS7960 motor driver is used to control high-power DC motors. It acts as an interface between the low-power control signals from the ESP32 and the high current requirements of the motors. The driver supports a high current capacity of up to 43A, making it suitable for driving high torque motors. It operates using PWM signals for speed control and an H-Bridge configuration for bidirectional control, allowing the robot to move forward and reverse. The driver also includes built-in protection features such as overcurrent and thermal protection, ensuring safe operation.

#### **5. Battery (LiPo)**

A Lithium Polymer (LiPo) battery is used as the power source for the robotic system. The selected battery has a nominal voltage of 11.1V and a capacity of 5200 mAh, which provides sufficient energy for continuous operation. The battery has a discharge rating of 40C, which means it can supply high current when required. This is essential for driving motors that draw large currents during startup and load conditions. The LiPo battery offers advantages such as high energy density, lightweight design, and stable output voltage.

### **10. Advantages of the System**

The proposed ultrasonic-based obstacle avoidance system offers several advantages in terms of performance, reliability, and scalability.

- **Real-time obstacle detection:**

The use of ultrasonic sensing and dedicated processing on Core 1 of ESP32 ensures immediate detection of obstacles, reducing the chances of collision.

- **Intelligent navigation using radar scanning:**

The integration of a servo motor allows the system to scan multiple directions and make informed decisions, rather than reacting only to front obstacles.

- **Efficient multitasking using FreeRTOS:**

The use of FreeRTOS enables parallel execution of multiple tasks such as sensing, scanning, and motor control, improving system efficiency and responsiveness.

- **High reliability and safety:**

The system incorporates threshold-based stopping, directional decision-making, and protected motor driving, ensuring safe operation even under varying conditions.

### 3.5 Robotics Arms:

#### 1. Introduction

The robotic arm system is integrated into the humanoid robot to perform human-like gestures such as waving, lifting, and directional movements. The arm mechanism is designed using four MG996R Servo Motor servo motors, controlled through Raspberry Pi 4 Model B GPIO pins using PWM signals. The system combines electrical interfacing, mechanical linkages, and software control to provide smooth and independent movement of both arms.

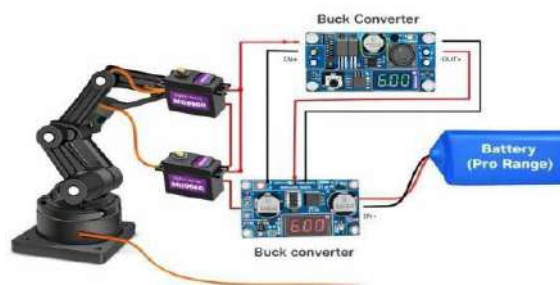
Each arm has **2 Degrees of Freedom (2-DOF)**:

1. Shoulder movement (up/down rotation)
2. Elbow movement (bending action)

Since two arms are used:

**Total DOF =  $2 \times 2 = 4$  DOF**

This arrangement allows the robot to perform gestures like greeting (“Hello” wave), arm raising, and coordinated arm m



**3.5.1 Circuit Diagram**

#### 2. Working Principle

The working principle of the robotic arm is based on converting electrical control signals into mechanical **movement**. The Raspberry Pi 4 Model B generates PWM (Pulse Width Modulation) signals and sends them to the MG996R Servo Motor servo motors. Depending on the pulse width, each servo rotates to a specific angle. This rotational motion is transferred through servo horns and aluminum brackets to move the shoulder and elbow joints of the robotic arm.

Power for the servos is supplied from the 11.1V Lithium Polymer Battery through the LM2596 Buck Converter, which regulates the voltage to 6V. The coordinated movement of four servos (two per arm)

produces human-like gestures such as lifting, bending, and waving. Thus, the system works on the principle of PWM-based servo control, mechanical linkage motion, and regulated power-driven actuation.

### **3. System Architecture and Sensor Arrangement**

The robotic arm system consists of four major sections:

#### **1. Power Supply Section**

The arm is powered using an 11.1V Lithium Polymer Battery battery.

- Battery output (11.1V) is supplied to the LM2596 Buck Converter
- LM2596 steps down voltage from 11.1V to regulated 6V
- This 6V is provided to all four servo motors
- Raspberry Pi is powered separately

This avoids overloading the Raspberry Pi.

#### **2. Control Section**

The Raspberry Pi 4 Model B acts as the main controller.

Functions:

- Generates PWM control signals
- Controls servo angle positions
- Executes Python program for gestures
- Receives trigger from robot main program
- Sends movement commands to arm motors

#### **3. Actuation Section**

Four MG996R Servo Motor servo motors convert electrical signals into mechanical rotation.

##### **Motor Arrangement**

##### **Right Arm**

- Servo 1 → Shoulder Joint
- Servo 2 → Elbow Joint

##### **Left Arm**

- Servo 3 → Shoulder Joint
- Servo 4 → Elbow Joint

##### **Torque Distribution**

- Shoulder servo bears full arm weight → highest torque requirement

- Elbow servo moves forearm → moderate torque requirement

#### **4. Mechanical Section**

The mechanical arm structure is made using aluminum brackets.

Components used:

##### **U-Shaped Bracket**

Acts as arm segment support.

Functions:

- Holds servo motor
- Transfers motion from servo horn
- Rotates arm segment

##### **Servo Horn**

Connects motor shaft to bracket.

Function:

- Transmits rotational force
- Converts motor shaft rotation into bracket movement

Without horn, arm will not move.

##### **L-Shaped Bracket**

Forms elbow joint.

Function:

- Mounts elbow servo perpendicular to shoulder servo
- Creates natural human-arm bending geometry

##### **Large U-Beam Servo Bracket**

Acts as forearm link.

Function:

- Extends arm length
- Connects elbow to end-effector/hand

#### **Interfacing with Raspberry Pi**

E & TC, AISSMS IOIT, Pune (Centre)

Each servo has 3 wires:

#### Servo Wire      Connection

Red (VCC)      6V from LM2596

Brown (GND)    Common Ground

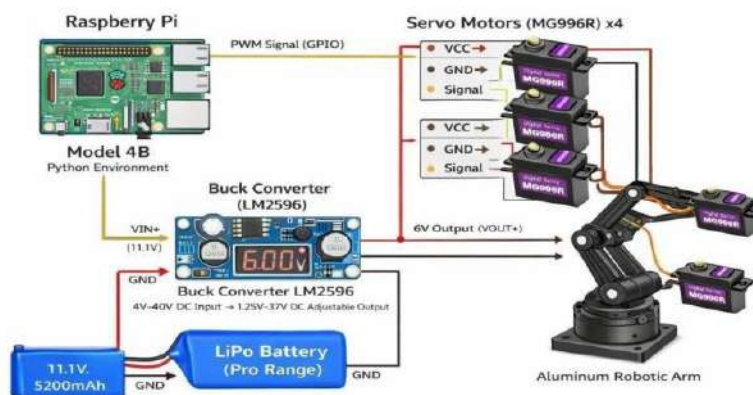
Orange (Signal) Raspberry Pi GPIO

Example:

- Signal → GPIO18
- Ground of battery, servo and Pi common

Common ground is mandatory for stable operation.

#### 4. Block diagram



#### 3.5.2 Robotic Arm Block Diagram

#### 5. Physical Dimensions of Robotic Arm

| Component        | Width (cm)    | Height (cm) | Weight (g) |
|------------------|---------------|-------------|------------|
| Single Arm       | 7.5 × 5       | 35          | —          |
| Clamps for Arm   | —             | 11.5        | 470        |
| MG996R Servo     | 1.97          | 4.29        | 55         |
| Shoulder Bracket | 4.5 × 3 × 4.5 | 9           | —          |
| Motor Bracket    | 3 × 4         | 7           | —          |
| L-Shaped Joint   | 6             | 2.5         | —          |

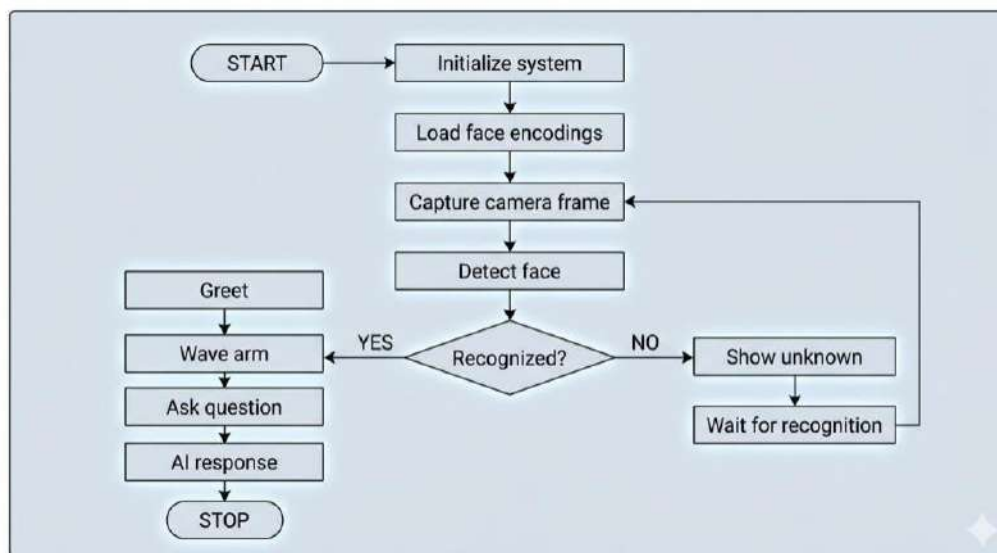
## 6. Working:

Electrically, the MG995R servo has three wires: VCC (Red), GND (Brown), and Signal (Orange). The signal wire connects directly to a GPIO pin on the Raspberry Pi (e.g., GPIO18). A common ground connection between the servo's GND, the Raspberry Pi's GND, and the battery's negative terminal is mandatory for stable signal transmission. The VCC wire is not connected to the Pi; instead, it connects to the output (6V) of the LM2596 buck converter.

For power management, the 11.1V LiPo battery is connected to the input of the LM2596 buck converter. Using a small screwdriver, the converter's potentiometer is adjusted to output exactly 6.0V (measured with a multimeter). This regulated 6V is then supplied directly to the servo's VCC wire. This approach isolates the high-current draw of the servo from the delicate 5V logic circuitry of the Raspberry Pi, preventing resets or permanent damage.

In software (Python), the RPi.GPIO library is used. The GPIO pin connected to the servo's signal wire is configured as a PWM (Pulse Width Modulation) output. The servo's position is controlled by varying the duty cycle of a 50Hz signal (period = 20ms). A pulse width of ~0.5ms (2.5% duty cycle) commands 0°, and ~2.5ms (12.5% duty cycle) commands 180°. To perform a "Hello" wave, the code would cycle the servo between two angles (0° and 90°) with a small delay, creating the back-and-forth waving motion. The trigger for this routine is received from the main loop (if person\_detected == True: perform\_wave()).

## 7. Algorithm



### 3.5.2 Robotic Arm Algorithm

## 8. Arm Integration

### 1. Introduction

- 2 arms  $\times$  2 DOF per arm = 4 servo motors total
- Face Detection - Finding faces in an image/video, locating where faces are in camera frame.
- Face Recognition - Identifying WHO the face belongs to, Matching detected face against known database.

The MG996R receives a PWM (Pulse Width Modulation) signal from the Raspberry Pi or ESP32. The width of the pulse determines the servo's shaft angle. A 0.5ms pulse commands 0°, a 1.5ms pulse commands 90°, and a 2.5ms pulse commands 180°. The internal control circuit drives a DC motor through the metal gear train until the output shaft reaches the commanded position.

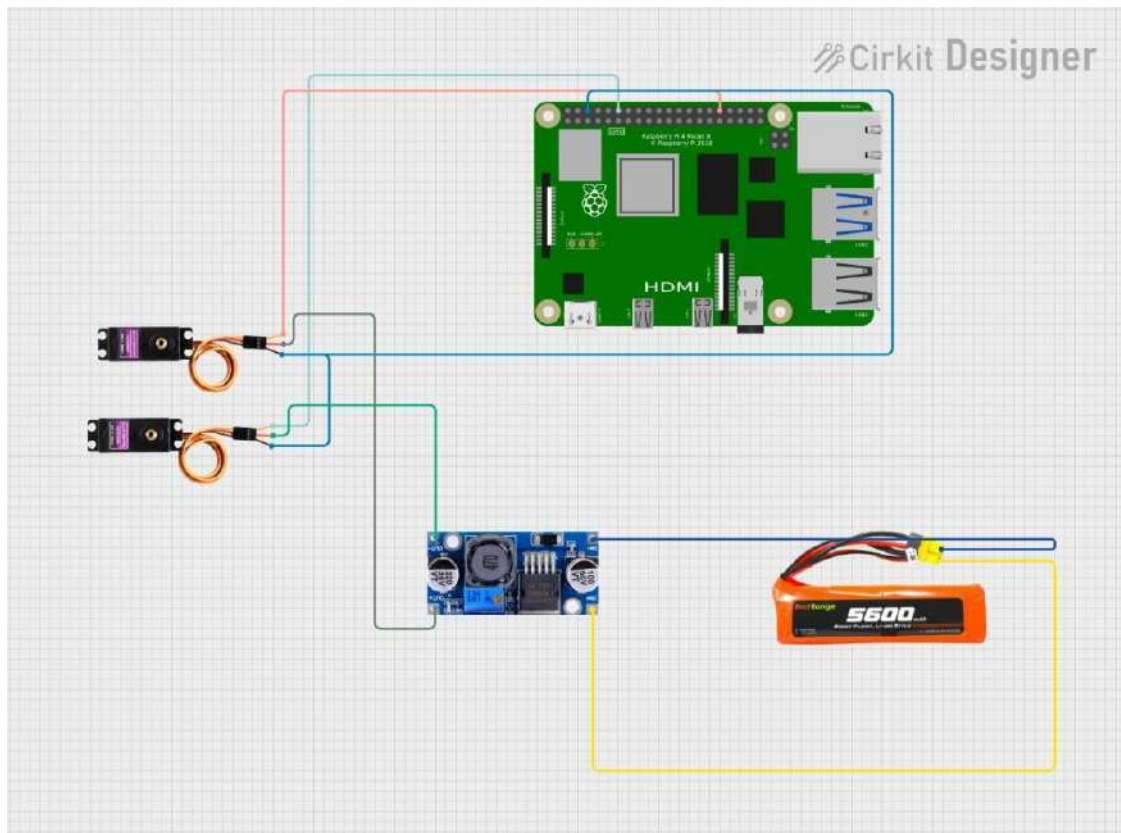
The servo horn is the connector that transmits rotational force from the servo's internal shaft to the external bracket. One horn is screwed directly onto the servo's output spline. The horn is then bolted to the bracket (U-shaped or L-shaped). Without the horn, the servo would spin freely without moving the arm. The U-shaped bracket is the structural backbone of each arm segment. The servo motor sits inside the U- channel. The servo horn, attached to the servo, bolts to the side walls of the U-bracket. When the servo rotates, the U-bracket rotates with it, moving the attached arm segment.

In Your 2-DOF Arm:

- First U-Bracket: Mounted on the shoulder servo (stationary base). This bracket holds the elbow servo.
- Second U-Bracket: Mounted on the elbow servo. This bracket extends to form the forearm.

Large U-Beam Servo Bracket is the forearm extension. While the standard U-bracket directly holds the servo, the "Large U-Beam" acts as the link between the elbow servo and the end of the arm. It provides the length needed to reach forward. The gripper or hand would attach to the far end of this large U-beam. The L-bracket acts as the "elbow joint connector". It is bolted to the rotating part of the shoulder U-bracket and holds the elbow servo motor. The 90-degree bend allows the elbow servo to be positioned perpendicular to the shoulder servo, creating the classic "human arm" geometry where the elbow bends in a different plane than the shoulder.

## 2. Block Diagram



### 3.5.3 Arm Integration Block Diagram

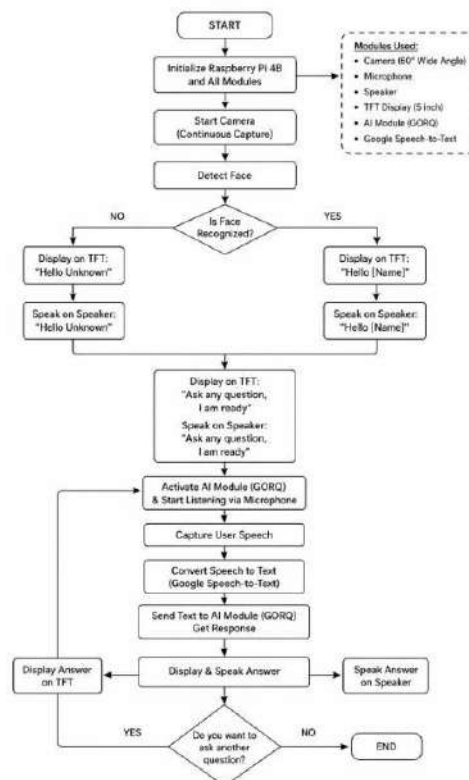
## 3.6 FACE RECOGNITION

### 1.Introduction

A smart face-recognition system based on a Raspberry Pi 4B. It uses a camera for face detection, a microphone for voice input, a speaker for voice output, and a 5-inch TFT screen for visual response. The system first detects a person's face, identifies whether the face is known or unknown, and then greets the person accordingly. After greeting, it activates the speech interface so the user can ask a question and receive an answer through both the speaker and the TFT display.

This combines computer vision, speech recognition, and AI-based response generation into a single compact embedded system. The face recognition part works on Raspberry Pi, while the question-answer part uses Google Speech-to-Text and the Groq AI module. This makes the system useful as a smart assistant.

### 2.System flow diagram



#### 3.6.1 Face Recognition

### **3. Working Principle**

1. It begins with continuous visual monitoring through the camera, which captures live images of the person standing in front of the robot. The captured image is processed to detect a face and compare it with the stored database of known faces. This recognition step allows the system to identify whether the user is a registered person or an unfamiliar person.

2. When a known face is detected, it displays a personalised greeting on the TFT screen in the form of “Hello [Name].” Along with the text, a ro

3. bo-eye facial expression is shown to make the interaction more natural and expressive. At the same time, the speaker produces an audio greeting so that the user receives both visual and voice feedback. If the face is not recognised, the system labels the person as unknown and displays “Hello Unknown” on the screen. This ensures that every detected face receives an immediate response from the system.

After the greeting stage, the system enters the interaction mode. In this mode, show the message “Ask any question, I am ready,” indicating that the system is ready to accept voice input. The microphone is then activated to capture the user’s speech. The spoken question is sent for speech-to-text conversion using Google Speech-to-Text technology.

4. The converted text is then forwarded to the Groq AI module. The AI analyses the question and generates a short, relevant response. In this way, the user receives the answer through both visual and audio channels, making the interaction clearer and more effective.

### **4. Hardware Interfacing**

The Raspberry Pi 4B acts as the main controller and communicates with all parts of the system. The camera is connected through the wired USB connection depending on the module used. The TFT screen is connected through HDMI for display and a B-type power input as required by the module. The microphone is connected through USB, while the speaker is connected through the 3.5 mm audio jack or USB audio output.

The Raspberry Pi 4 Model B has 2 × micro-HDMI ports, 2 × USB 2.0 ports, 2 × USB 3.0 ports, a CSI camera port, a DSI display port, a 3.5 mm audio jack, and USB-C power input. Its power requirement is 5v.

## Display Output

The output display is one of the most attractive parts of this project. The TFT shows different screen states such as waiting mode, welcome mode, unknown-person mode, listening mode, thinking mode, and answer mode. In greeting mode, the screen displays the face name and a friendly robot expression. In unknown mode, it shows that the face is not recognized. During question answering, it displays a short AI response in a readable format.

This visual communication improves interaction and makes the system easy to understand even for a non-technical user. The combination of text, animation-like face expression, and spoken response creates a complete human-machine interface.

## Libraries Used

The following Python libraries support the mechanism of system: face\_recognition for face detection and encoding, cv2 for camera frame handling, numpy for data processing, pygame for TFT graphics, RPi.GPIO for hardware control, speech\_recognition for voice capture and Google Speech-to-Text integration, gtts for speech generation, and groq for AI response generation. These libraries together allow the Raspberry Pi to handle vision, audio, display, and AI functions in one program.

The code also uses threading, queue, subprocess, pickle, csv, and time to manage background tasks, store face data, log detections, and keep the interface responsive. This is important because face recognition, audio playback, and AI response generation need to run without blocking each other.

## 5. Advantages of the System

The proposed AI-based interactive system offers several advantages in terms of performance, usability, and scalability.

- **Real-time interaction and quick response generation:**

The integration of speech recognition and AI processing enables the system to respond instantly to user queries, ensuring smooth and efficient communication.

- **Dual output system (display + audio) improves usability:**

The combination of TFT display and speaker provides both visual and audio feedback, making the system more interactive and user-friendly.

- **Easy integration of multiple modules (camera, mic, display):**

The system supports seamless interfacing of camera, microphone, and display modules, allowing efficient coordination between hardware components.

- **Scalable for future upgrades like IoT or automation:**

The design allows easy expansion to advanced applications such as smart home systems, IoT connectivity, and automation features.

- **Low power consumption suitable for continuous operation:**

The use of efficient embedded hardware ensures minimal power usage, making the system ideal for long-term and continuous use.

## **6. Component Specification**

### **1. Raspberry Pi 4B**

The Raspberry Pi 4B is the main controller of the project. It has a 40-pin GPIO header, HDMI support, CSI camera port, USB ports, a 3.5 mm audio jack, and a 5 V at 3 A power input. It processes face detection, controls the TFT display, and manages the microphone and speaker. It also runs the AI and speech recognition software, so it works as the brain of the system.

### **2. Camera**

The camera is used to capture the face of the person in front of the system. It has a 60 degree wide-angle lens, wired connection, 61 mm length, and 32 mm width. It sends live images to the Raspberry Pi for face recognition. It is the first input device in the project.

### **3. Speaker**

The speaker is used for voice output in the project. It can be connected through a 3.5 mm jack or USB. It speaks greetings, unknown-face alerts, and AI answers. It helps make the system interactive and easy to use.

### **4. TFT Display**

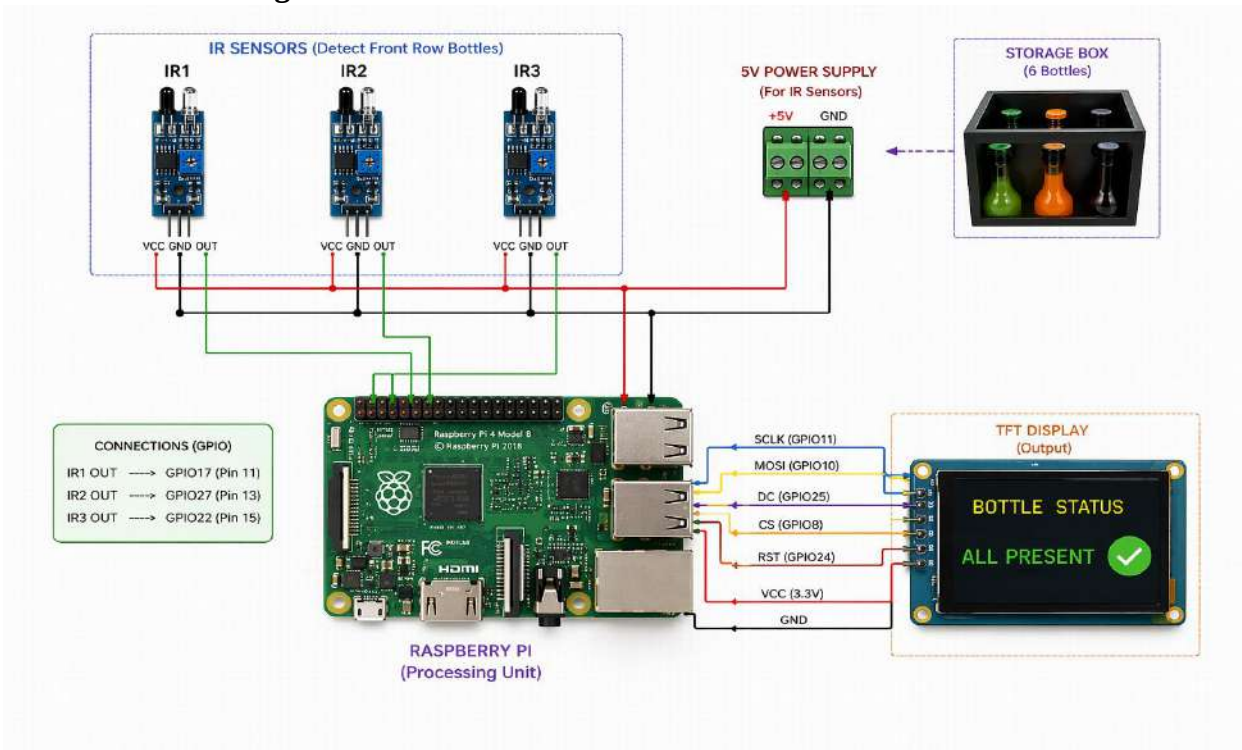
The TFT display shows the visual output of the system. It is a 5-inch display with an HDMI port and B-type charging port. It displays the face name, unknown message, listening status, and AI response. It also shows robo-eye expressions to make the system look friendly.

### **5. Microphone**

The microphone is used to take voice input from the user. It has a frequency response of 100 to 16 kHz, a 90 cm cable length, and a 13 cm microphone length. It captures the user's question and sends it for speech-to-text conversion. It is an important part of the voice interaction stage.

## 3.7 Bottle Detection System Using IR Sensors

### 1. Circuit Diagram



### 3.7.1 IR Sensors Diagram

## 3. Introduction

The bottle detection system is a crucial subsystem of the serving robot, responsible for monitoring the availability of bottles placed inside a storage box. This system ensures that the robot is aware of whether bottles are present, partially removed, or completely absent, enabling proper decision-making during operation.

In this project, the bottle detection mechanism is implemented using three infrared (IR) sensors interfaced with a Raspberry Pi. The storage unit consists of six bottles arranged in two rows, namely the front row and the back row. The IR sensors are mounted strategically on the front side of the storage box to detect only the front row bottles.

The Raspberry Pi acts as the processing unit, continuously reading sensor inputs through its GPIO pins and determining the bottle status using predefined logic. The processed result is then displayed on a TFT display, providing real-time feedback to the user.

This approach reduces hardware complexity and cost while maintaining sufficient accuracy for the intended application.

## 4. Working Principle of IR Sensor System

The IR sensor operates based on the principle of infrared radiation and reflection. Each sensor module consists of two main components: an IR transmitter (IR LED) and an IR receiver (photodiode or phototransistor).

The working mechanism is explained as follows:

- The IR transmitter continuously emits infrared radiation.
- When a bottle is present in front of the sensor, the emitted IR rays are reflected back.
- The IR receiver detects this reflected radiation and converts it into an electrical signal.
- The sensor module processes this signal and provides a digital output.

Output Characteristics:

- Logic HIGH (1) → Bottle is present (reflection detected)
- Logic LOW (0) → Bottle is absent (no reflection)

The Raspberry Pi reads these digital outputs and uses them to determine the overall bottle status.

## 3. System Architecture and Sensor Arrangement

The bottle detection system is designed with an optimized sensor arrangement to minimize hardware usage while maintaining functionality.

- Total bottles: 6
- Arrangement: 2 rows (Front + Back)
- Sensors used: 3 IR sensors

### Sensor Placement:

| Sensor | Position     | Function         |
|--------|--------------|------------------|
| IR1    | Front left   | Detects Bottle 1 |
| IR2    | Front Center | Detects Bottle 2 |
| IR3    | Front Right  | Detects Bottle 3 |

The back row bottles are not directly detected. Instead, the system assumes their presence based on the front row detection.

#### 4. Detection Logic and Decision Making

The Raspberry Pi processes the sensor inputs using logical conditions to determine bottle availability.

##### Case 1: All Bottles Present

When all sensors detect bottles:

- IR1 = HIGH
- IR2 = HIGH
- IR3 = HIGH

→ System Output: **ALL BOTTLES PRESENT**

This condition indicates that all front bottles are present, and by assumption, all six bottles are available.

##### Case 2: Bottle Removed

When one or more sensors do not detect a bottle:

- Any sensor = LOW

→ System Output: **BOTTLE REMOVED**

This indicates that at least one bottle from the front row has been removed.

##### Case 3: All Bottles Absent

When no sensor detects a bottle:

- IR1 = LOW
- IR2 = LOW
- IR3 = LOW

→ System Output: **ALL BOTTLES ABSENT**

This indicates that all front bottles are missing, and hence the system assumes all bottles are removed.

#### 5. Interfacing with Raspberry Pi

The IR sensors are interfaced with the Raspberry Pi using GPIO pins for digital input.

##### GPIO Connections:

- IR1 OUT → GPIO17 (Pin 11)
- IR2 OUT → GPIO27 (Pin 13)
- IR3 OUT → GPIO22 (Pin 15)

##### Power Supply Configuration:

To simplify the circuit design, a **common power configuration** is used:

- All sensor VCC pins are connected to a common +5V supply
- All sensor GND pins are connected to a common ground

This reduces wiring complexity and improves system reliability.

#### 6. Output Display System (TFT Display)

The system uses a TFT display to provide real-time visual feedback regarding bottle status. The Raspberry Pi processes sensor data and sends corresponding messages to the display.

**Display Outputs:**

- “ALL PRESENT” → All bottles available
- “BOTTLE REMOVED” → At least one bottle missing
- “ALL ABSENT” → No bottles present

The TFT display enhances user interaction and provides clear system status.

## **7. Flowchart Description**

The system operation follows a continuous loop-based process:

1. Initialize Raspberry Pi GPIO pins
2. Configure IR sensors as input devices
3. Read sensor values (S1, S2, S3)
4. Apply decision logic:
  - If all inputs are HIGH → Display “ALL PRESENT”
  - If all inputs are LOW → Display “ALL ABSENT”
  - Otherwise → Display “BOTTLE REMOVED”
5. Update TFT display with result
6. Wait for a short delay
7. Repeat the process continuously

## 3.8 AI Module

### 1. Use of AI Module in the Project:

The Artificial Intelligence (AI) module plays a crucial role in enhancing the interaction capability of the robot by enabling intelligent and human-like communication. In this project, the AI module is implemented using the Groq API, which utilizes the LLaMA 3.1 language model to process natural language queries and generate meaningful responses.

Where the AI Module is Used:

The AI module is activated after the completion of the face recognition and greeting process. Once the robot successfully identifies a person and greets them using the TFT display, robotic arm gesture, and Text-to-Speech output, it enters the interaction phase. During this phase:

1. The system prompts the user to ask a question (displayed on TFT screen and spoken via TTS).
2. The user's voice input is captured through a microphone.
3. The audio is converted into text using Google Speech-to-Text (STT).
4. This recognized text is then sent as input to the AI module (Groq API).
5. The AI processes the input and generates an appropriate response.
6. The response is then converted into speech using Google Text-to-Speech (TTS) and displayed on the TFT screen.

Thus, the AI module acts as the core decision-making and conversational engine during the human-robot interaction stage.

Why the AI Module is Used:

The AI module is used in this project for several important reasons:

1. **Natural Language Understanding**  
The AI module allows the robot to understand human language in a flexible way rather than relying on fixed commands. This makes interaction more natural and user-friendly.
2. **Intelligent Response Generation**  
Instead of pre-programmed replies, the AI generates dynamic and context-aware responses, making the robot capable of handling a wide range of questions.

3. Improved User Experience

By integrating AI, the robot behaves more like a real assistant rather than a simple machine, which enhances engagement and usability in environments like reception areas or homes.

4. Scalability and Flexibility

The use of a powerful language model like LLaMA allows the system to be easily upgraded with more advanced conversational capabilities without major hardware changes.

5. Real-Time Interaction

The Groq API provides fast processing, enabling near real-time responses, which is essential for smooth conversation

# CHAPTER 4: RESULTS AND DISCUSSION

## 4.1 Results

The Guest-Greeting and Beverage-Serving Robot was successfully designed, assembled, and tested. The following outcomes were observed during the testing phase:

- 1. Navigation & Line Following** The robot successfully followed the predefined black line path using IR sensors with consistent accuracy. It moved smoothly across the test surface without deviation under normal lighting conditions.
- 2. Obstacle Detection** The ultrasonic sensor (HC-SR04) effectively detected obstacles within a 20–30 cm range and triggered the robot to stop, preventing any collisions during operation.
- 3. Beverage Delivery** The robot successfully carried 6 water bottles in its enclosed storage compartment and delivered them to the designated endpoint along the line path without any spillage or instability.
- 4. Guest Greeting** The TFT display showed welcome messages and the gTTS-powered speaker delivered voice greetings successfully upon reaching the guest location, creating an interactive dual-mode greeting experience.
- 5. Robotic Arm Gesture** The servo-based robotic arm performed the "Hi" wave gesture as programmed, adding a human-like and friendly interaction element to the robot's behavior.
- 6. Power Performance** The 11.1V 5200mAh battery powered all subsystems consistently throughout the testing duration with no significant voltage drops observed.

## 4.2 Problems Faced

- 1. Line Detection Issues** IR sensors were sensitive to ambient lighting conditions, causing occasional deviation from the line path under bright or uneven lighting.
- 2. Servo Motor Vibration** Servo motors produced slight vibrations during movement, causing minor instability in the robot's body and the storage compartment holding the bottles.
- 3. Power Distribution** Managing proper voltage levels for Raspberry Pi (5V) and servo motors (6V) from the 11.1V battery required careful use of step-down converters to avoid component damage.
- 4. Robotic Arm Calibration** Fine-tuning the servo angles for a smooth and natural "Hi" wave gesture required multiple iterations of PWM value adjustments.

**5. Audio Delay** A slight delay was observed in gTTS audio playback due to internet dependency for speech generation, causing a minor lag in the greeting response.

**6. Weight Distribution** Placing 6 water bottles in the compartment slightly affected the robot's balance and speed, requiring adjustments in motor control logic.

**7. TFT Visibility** The display was not clearly visible in bright environments, requiring contrast and positioning adjustments for better readability.

## 4.3 Hardware Results

**Chassis & Body** The robot body was successfully assembled with a sturdy lightweight frame, capable of holding all components and 6 water bottles without structural instability.

1. **Servo Motors (Locomotion)** Servo motors drove the wheels smoothly along the line path, responding accurately to IR sensor inputs for directional corrections.

2. **IR Sensors** The IR sensor array successfully detected the black line and sent real-time signals to Raspberry Pi for navigation control with good response time.

3. **Ultrasonic Sensor** HC-SR04 successfully measured distances and triggered stop signals when obstacles were detected within the 20–30 cm safety range.

4. **Robotic Arm** The servo-based arm was physically mounted and performed the "Hi" wave gesture with smooth angular movement across the defined range.

5. **TFT Display** The TFT module displayed welcome messages correctly upon reaching the guest location, functioning as expected.

6. **Speaker Module** The speaker successfully played gTTS-generated voice greetings with clear audio output during testing.

7. **Battery** The 11.1V 5200mAh battery powered all hardware components simultaneously without any power failure or overheating during the test run.

8. **microphone** Successfully captured guest voice input during testing. The microphone responded accurately to voice commands in indoor environments with low background noise. Minor interference was observed in outdoor testing due to ambient noise, requiring slight sensitivity adjustments

## 4.4 Difference Between Simulation and Hardware:

| Parameter          | Simulation                            | Hardware                                              |
|--------------------|---------------------------------------|-------------------------------------------------------|
| Environment        | Ideal, controlled virtual environment | Real-world with physical constraints                  |
| Line Following     | Perfect path tracking without errors  | Minor deviations due to surface texture and lighting  |
| Obstacle Detection | Instant and accurate response         | Slight delay due to sensor response time              |
| Servo Motors       | Smooth and precise movement           | Minor vibrations and mechanical play observed         |
| Robotic Arm        | Exact angular movement as programmed  | Required fine-tuning of PWM values for smooth gesture |
| Audio Greeting     | Immediate playback without delay      | Slight delay due to gTTS internet dependency          |
| Power Supply       | No power limitations                  | Voltage regulation required for multiple components   |
| Weight & Balance   | No physical weight impact             | Bottle load affected speed and stability              |
| LCD Display        | Always clear and visible              | Visibility affected in bright lighting conditions     |
| Cost               | Free / No cost                        | Component and assembly cost involved                  |
| Testing            | Fast, repeatable, risk-free           | Time-consuming with risk of hardware damage           |
| Accuracy           | 100% as per code logic                | ~85–90% due to real-world interference                |

## 4.5 Comments & Reasons

- 1. Line Following Deviation Comment:** Robot occasionally deviated from the line path. **Reason:** IR sensors were affected by ambient light and surface reflectivity variations.
- 2. Servo Vibration Comment:** Slight vibration observed during movement. **Reason:** PWM signal inconsistencies and mechanical play in servo gears caused minor instability.
- 3. Audio Delay Comment:** Voice greeting had a slight delay. **Reason:** gTTS requires an active internet connection to generate speech, causing network-dependent latency.
- 4. Arm Calibration Comment:** Robotic arm required multiple adjustments. **Reason:** Default PWM values did not produce natural gesture angles, requiring manual fine-tuning.
- 5. Weight Imbalance Comment:** Robot slowed down when fully loaded with bottles. **Reason:** Added weight of 6 bottles shifted the center of gravity, affecting motor torque and speed.
- 6. Visibility Comment:** Display was unclear in bright environments. **Reason:** backlight intensity was insufficient against strong ambient or sunlight.
- 7. Power Management Comment:** Voltage fluctuations observed during peak load. **Reason:** Multiple servo motors running simultaneously drew high current, causing brief voltage dips.
- 8. Overall Performance Comment:** Robot successfully completed all intended tasks. **Reason:** Proper integration of all components with well-structured Python code ensured smooth overall functioning.

# CHAPTER 5: CONCLUSION & FUTURE SCOPE

## 5.1 Conclusion

The Guest-Greeting and Beverage-Serving Robot was successfully designed and developed as a low-cost, efficient, and interactive service robot. The robot demonstrated reliable performance in all core functionalities — line following navigation, obstacle avoidance, beverage delivery, voice and display-based guest greeting, and human-like "Hi" wave gesture through the servo-based robotic arm.

The project proved that affordable components like Raspberry Pi, servo motors, IR sensors, and gTTS can be effectively combined to build a functional service robot suitable for real-world hospitality environments such as offices, colleges, hospitals, and events.

Despite minor challenges such as IR sensor sensitivity, audio delay, and servo calibration, all issues were identified, analyzed, and resolved during the testing phase, resulting in a stable and functional prototype.

The project successfully achieved its objectives and serves as a strong foundation for future enhancements such as face recognition, wireless control, multi-gesture robotic arm, and AI-based autonomous navigation, making it scalable for more advanced service robot applications

## 5.2 Future Scope

1. Face Recognition Integrating OpenCV-based face recognition to identify and greet guests by name, providing a more personalized experience.
2. Wireless Control Adding Wi-Fi or Bluetooth-based remote control for manual override and monitoring of the robot from a smartphone or laptop.
3. AI-Based Navigation Replacing line following with AI-powered autonomous navigation using camera and deep learning for dynamic path planning in unstructured environments.
4. Multiple Arm Gestures Expanding the robotic arm's capabilities to perform multiple gestures such as handshake simulation, pointing, and namaste for culturally diverse interactions.
5. Voice Command Recognition Incorporating speech recognition to allow guests to interact with and give commands to the robot verbally.
6. Multi-Language Support Extending gTTS greeting capability to support regional languages like Hindi, Marathi, and others for wider accessibility.
7. Automated Charging Implementing a self-docking charging station so the robot autonomously returns and recharges when battery is low.
8. IoT Integration Connecting the robot to an IoT platform for real-time monitoring, remote management, and data logging of robot activity.
9. Enhanced Payload Redesigning the storage compartment to carry additional items like snacks, documents, or medicines for broader use cases.
10. Commercial Deployment Scaling the prototype into a market-ready product for hotels, hospitals, and corporate offices with improved build quality and reliability.

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## APPENDIX



